

AI+RES Event Slate

Nine CONUS cases put through rare-event resampling at week-3 lead, from a heat dome no direct ensemble reaches to an ordinary Friday in August. Each one gets its observed anomaly, how rare that was against 64 years of ERA5, and the probability the run assigns to it.

Every number on this page reduces to one observable:

A_L = $\cos(\text{lat})$ -weighted mean, over the event box, of the 7-day-mean 2 m temperature anomaly ending at the event peak

GenCast is initialised 21 days before that peak and rolled as 64 walkers. At five barriers FourCastNet 3 scores each walker's expected A_L , the promising ones are cloned and the rest killed, and a weight w_i undoes the tilt so the surviving population still estimates GenCast's own probabilities. The C schedule (0, 1.0, 1.4, 1.8, 2.0) was tuned once, on the Pacific Northwest, and has not been re-tuned for any event since.

The severity ladder

Ordered by how far the observation sat outside GenCast's own week-3 forecast distribution, because that - not the size of the anomaly, and not its climatological rarity - is what predicts whether resampling reaches it. Winter Storm Elliott, at -14.42 K, is reached; the Southwest heat wave at $+5.14$ K is not, because it sits half a sigma deeper in the model's spread.

RUNG	EVENT	REGIME	OBSERVED A_L	CLIM. PCT	RETURN PERIOD	Σ - DEPTH	DIRECT	AI+RES	WEIGHTED P	OUTCOME
1	Southwest Heat Wave August 2020	heat	+5.14 K	99.96th	210 yr	+3.3 σ	0 / 24	0 / 64	$< 3 \times 10^{-4}$	not reached
2	California Heat Wave September 2022	heat	+4.71 K	99.71th	37 yr	$\sim +2.9$ σ	0 / 24	25 / 64	0.0186	reached
3	Winter Storm Elliott December 2022	cold	-14.42 K	98.41th	5 yr	+2.81 σ	0 / 24	20 / 64	0.0079	reached
4	Winter Storm Uri February 2021	cold	-7.40 K	96.80th	4 yr	+2.28 σ	0 / 24	32 / 64	0.0090	reached
5	South-Central Heat Dome June 2023	heat	+5.73 K	100.00th	beyond the record	$\sim +2.2$ σ	0 / 16	6 / 64	0.00273	reached

RUNG	EVENT	REGIME	OBSERVED A _L	CLIM. PCT	RETURN PERIOD	Σ- DEPTH	DIRECT	AI+RES	WEIGHTED P	OUTCOME
6	Pacific Northwest Heat Dome June 2021	heat	+7.72 K	99.60th	22 yr	+1.98 σ	0 / 40	42 / 64	0.054	reached
7	The Nationwide Warm Spell 24 December 2025	CONUS index	+3.26 K	98.78th	10 yr	+1.91 σ	1 / 24	46 / 64	0.0661	anchor
8	The Normal-Day Test 2 August 2024	CONUS index	+1.06 K	94.08th	4 yr	0.00 σ	14 / 24	63 / 64	0.649	control
9	The Below-Median Day 7 November 2023	CONUS index	+0.21 K	60.10th	1 yr	-0.97 σ	21 / 24	64 / 64	0.583 (pinned)	pinned

Two different questions live in the two percentile columns and they answer differently on purpose. The **climatological percentile** is how rare the weather was: the same index computed from 64 years of ERA5, pooled within ±21 days of the event's calendar position. **σ-depth** is how hard it was to forecast: where the observation sat in the 24-member GenCast ensemble launched three weeks earlier. The South-Central heat dome is the rarest case here and only the fifth-hardest to reach.

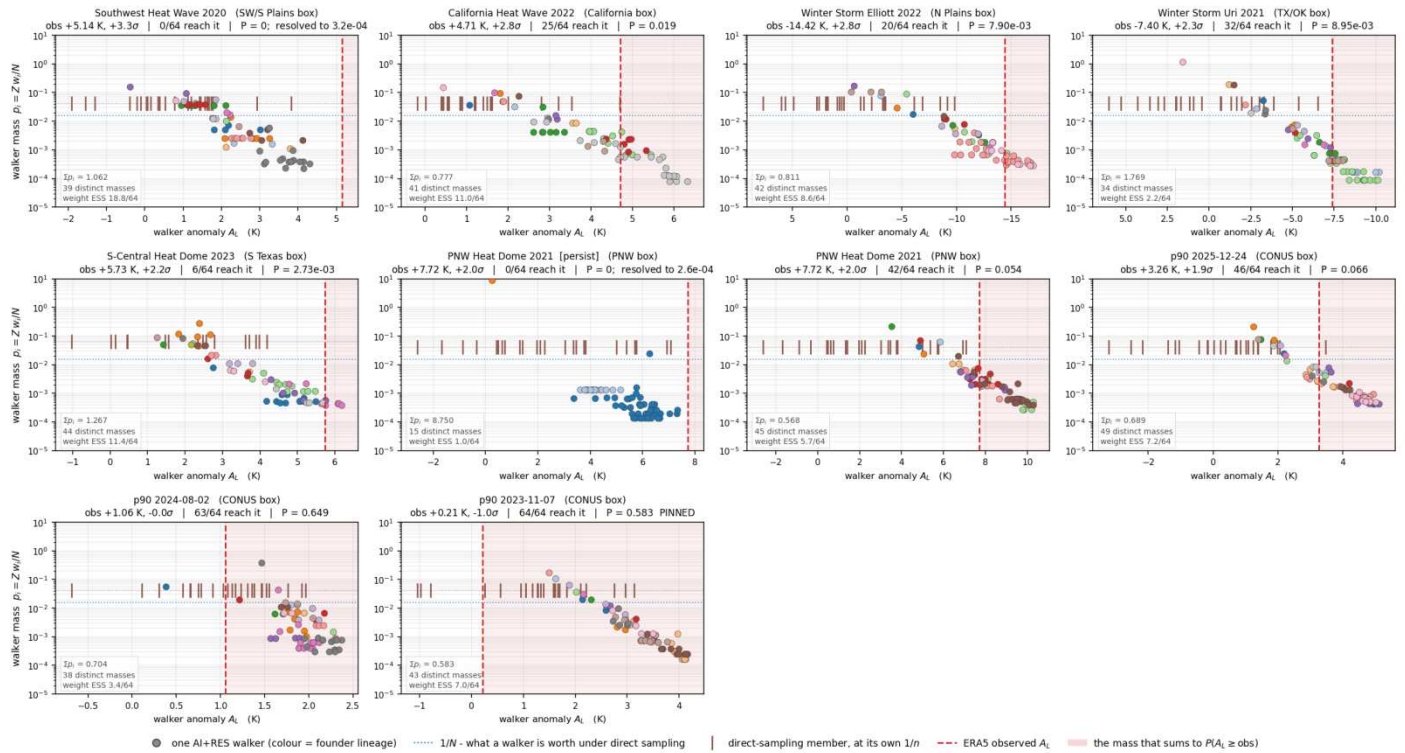
The whole population

all ten runs CROSS-EVENT

Every number in the table above is a sum over the walkers of one run. This is that raw material, one panel per run, in the same hardest-first order: where each of the 64 surviving walkers landed on A_L , and the probability mass $p_i = Z \cdot w_i / N$ it carries. The shaded band is the mass that sums to the headline P .

Three things are legible here that a table cannot show. **Reach is bought and then charged for:** the blue line is $1/N$, what a walker would be worth unresampled, and the deepest walkers sit one to two decades below it - the Pacific Northwest's hottest carries 2.5 % of what a single direct member is worth. **The masses come in tied blocks:** walkers cloned at the last barrier inherit their parent's mass exactly, so Uri's 64 walkers hold 34 distinct values and the persistence control's hold 15, and that count - not 64 - is the number of independent pieces of information behind the estimate. **Two runs cannot answer their own question:** Southwest and the control have nothing in the band, so P is 0 and the honest statement is the deepest level the run resolved (3.2 and 2.6×10^{-4}); 7 November 2023 has the opposite failure, its observation below every walker, so the band covers the whole panel and P is just the run's own normalization check.

AI+RES: every surviving walker, its anomaly, and the probability it carries
each panel's headline P is the mass right of the red line; $\sum p_i$ over the whole panel is the run's normalization check



Panels ordered by how far the observation sat outside the DIRECT ensemble's spread, hardest first. A walker far below the 1/N line is one resampling manufactured and the weights are charging for; walkers stacked at one height are clones of a single parent at the last resampling and carry identical mass by construction. Two runs cannot answer their own question: Southwest and the persistence control reached nothing, so P is 0 and the honest statement is the deepest level the run resolved, and p90 2023-11-07 is PINNED - the observation lies below every walker, so the subset sum is the whole panel and P equals the normalization check, not a probability.

The population, all ten runs. One panel per run, ordered by how far the observation sat outside the direct ensemble's spread. Each dot is a walker, coloured by founder lineage; the brown ticks are the direct-sampling members at their own $1/n$; the red dashed line is the observation and the shaded side of it is the mass that sums to P . $\sum p_i$ over a whole panel is that run's normalization check, which should be 1 and is quoted in each panel's corner alongside the weight ESS.

The trajectory panels below now also carry the operational CFSv2 forecast at the same 21-day lead, on all nine events rather than the original four. It reaches the observation only on the two mildest rungs - 3 of 4 members on 7 November 2023, 2 of 4 on 2 August 2024 - and misses every genuinely extreme one, including the 24 December 2025 warm spell by 4.6 K with its ensemble mean on the wrong side of climatology. That is not a resolution artefact: pushing the ERA5 truth field through CFS's own $\sim 0.94^\circ$ grid and re-scoring it moves A_L by at most 0.13 K, against gaps of 2.3 to 11.6 K.

Southwest Heat Wave

August 2020

RUNG 1 OF 9

heat not reached

The honest miss. The deepest target relative to GenCast's own spread, and the one run that does not reach it - the observation is $+3.3 \sigma$ out, where a $C \leq 2$, $K = 5$ tilt moves the population mean about $+1.75 K$, not $+4.4 K$.

OBSERVED A_L

+5.14 K ERA5, 29-35 N, 109-103 W

CLIMATOLOGICAL PERCENTILE

99.96th ERA5 1959-2022 · 100.00th against 1990-2019

RETURN PERIOD

210 yr 1 of 64 - itself years hold a window this extreme

DEPTH IN GENCAST'S SPREAD

+3.3 σ week-3 ensemble $+0.71 \pm 1.33 K$

DIRECT SAMPLING

0 / 24 best member $+3.82 K$ · $P = 0.000$ [0.000, 0.142]

AI+RES

0 / 64 most extreme $+4.31 K$

WEIGHTED $P(A_L \text{ BEYOND OBSERVED})$

$< 3 \times 10^{-4}$ sum of the walker probabilities past the observation

PEAK-DAY ANOMALY

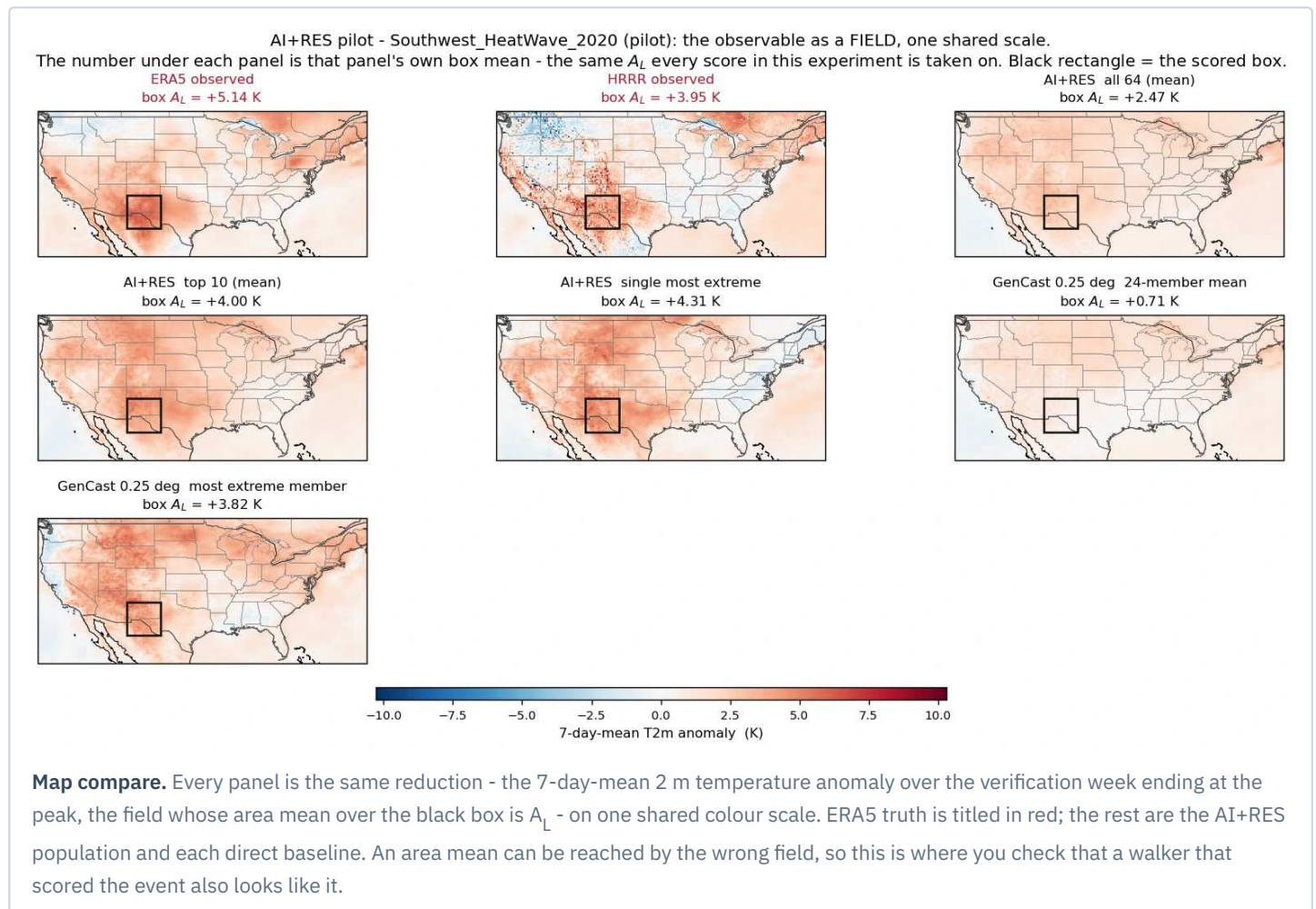
+4.16 K the single day, not the week · 99.1th percentile

64-YEAR RECORD FOR THIS SEASON

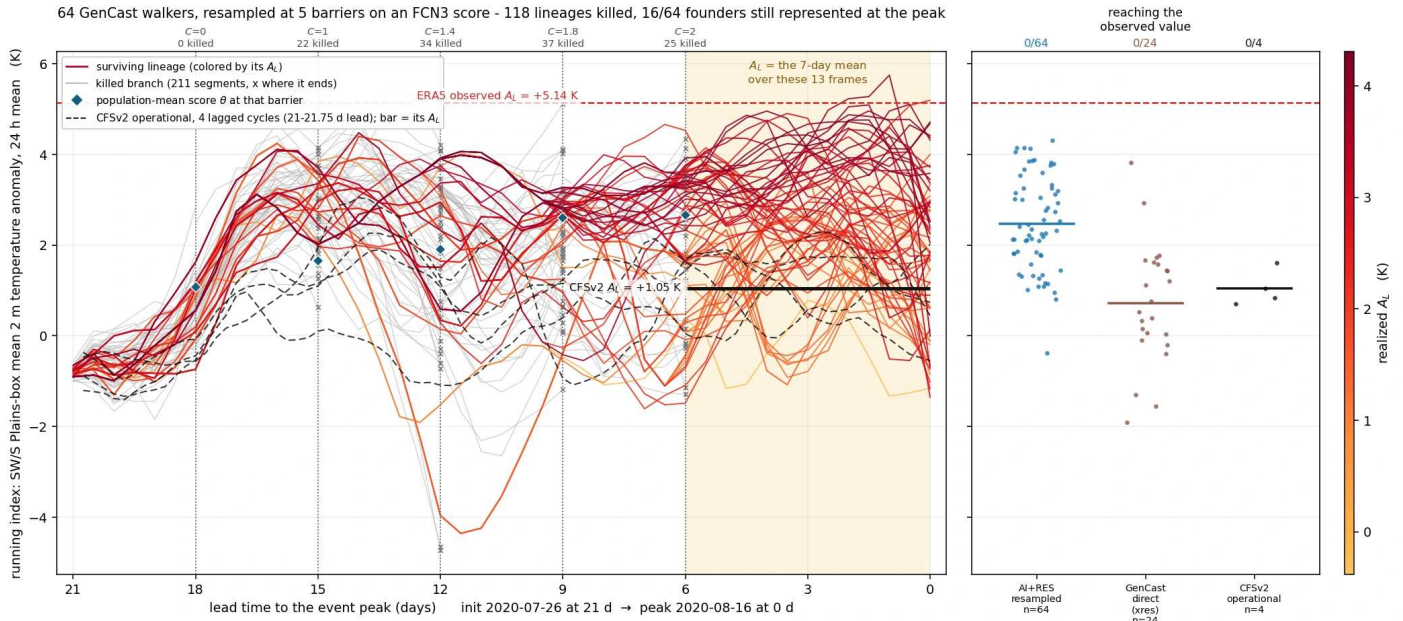
+5.15 K - this event peak 2020-08-16 · init 2020-07-26 (21 d lead)

box: Southern New Mexico and west Texas (29-35 N, 109-103 W) · registry key
 Southwest_HeatWave_2020 · Slurm job 1176, 7 h 34 m
 run health: log Z -0.8660 · normalization check 1.0623 · 16/64 founders · ESS/N crashed to 0.11 at
 step 3 (max multiplicity 23), recovering to 0.25 and 0.37

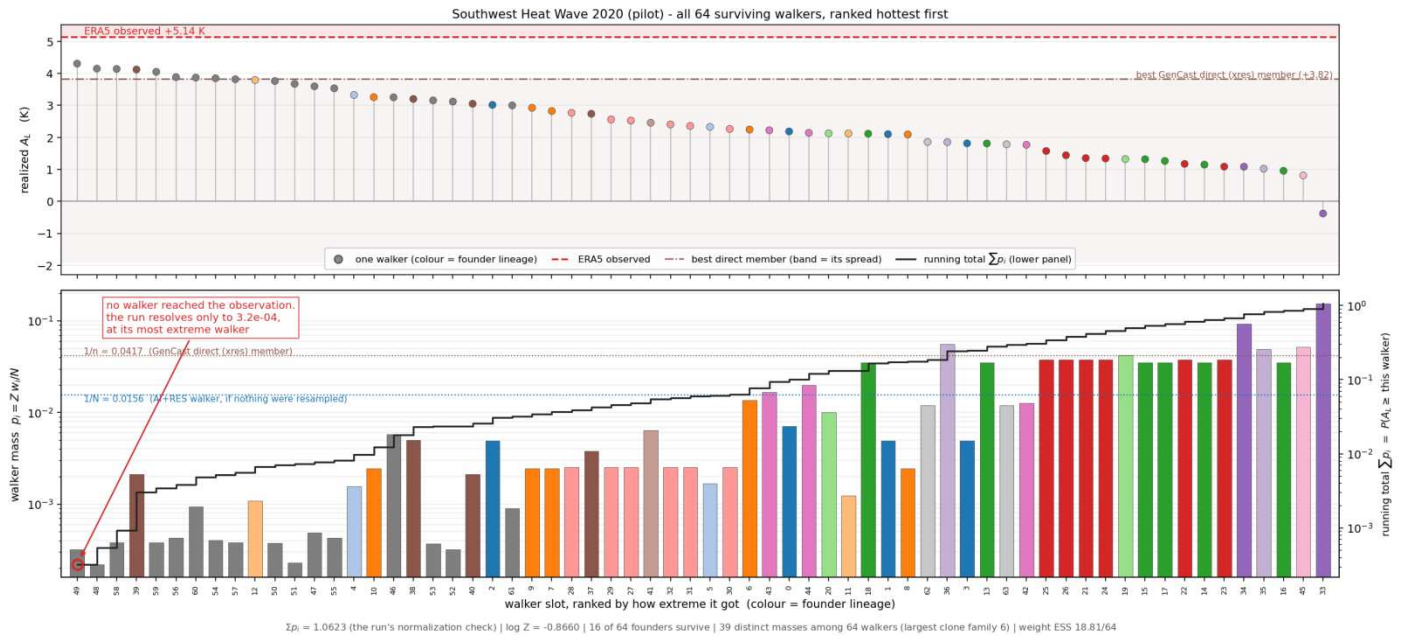
What makes the miss worth as much as the hits is that the curve is still honest where it can be checked:
 through the range 24 direct members resolve, the weighted exceedance tracks direct sampling inside
 binomial noise ($P(\geq +2.23 \text{ K})$ 0.077 against 0.083; $P(\geq +3.03 \text{ K})$ 0.026 against 0.042), then extends
 about two decades deeper. The weights deflate the tilt as they should - weighted mean +1.24 K against
 an unweighted +2.47 K.



AI+RES pilot - Southwest_HeatWave_2020 (pilot): the walk itself, barrier by barrier



Trajectory. The algorithm happening: A_L against lead time for every branch the run rolled, grey for lineages that were killed and coloured for the 64 that survived, with the resampling barriers as vertical lines, the 7-day verification window shaded, the observed value in red, and the operational CFSv2 forecast at the same lead in black.

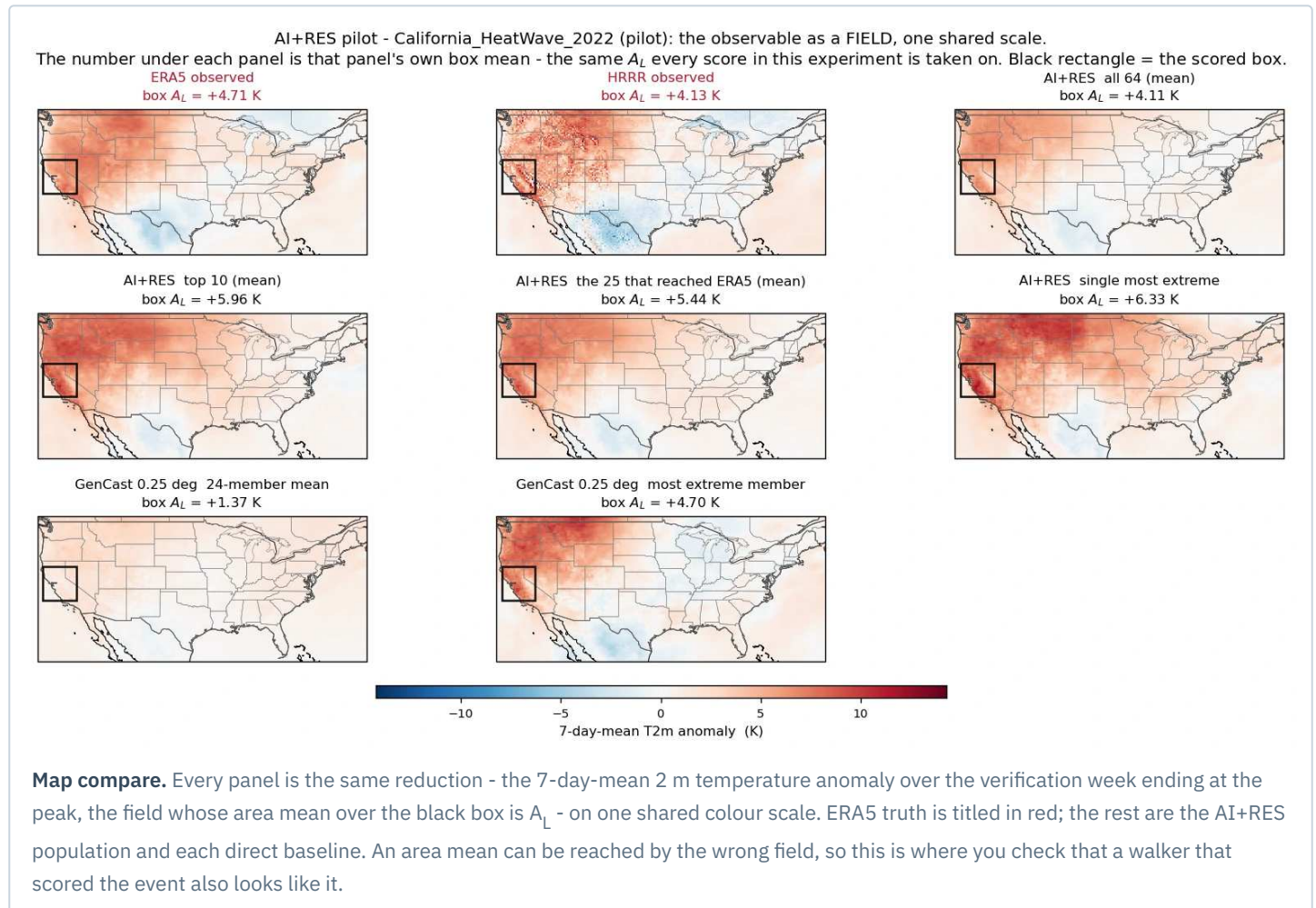


Every walker, and what it is worth. Top: all 64 surviving walkers ranked by how extreme they got, against the observed value in red and the best direct-sampling member in brown. Bottom: the probability mass $p_i = Z \cdot w_i / N$ each one carries, on a log axis, with the running total $\sum p_i$ as the black staircase - the headline probability is where that staircase stands when the ranking crosses the observation. Two reference lines set the scale: $1/n$, what a single direct member is worth, and $1/N$, what a walker would be worth if nothing had been resampled. Bars at identical height are clones of one parent at the last resampling and carry identical mass by construction, so the count of distinct masses under the panel - not 64 - is how many independent pieces of information the estimate rests on.

A hit at moderate severity. Direct sampling's best member grazed the observation at +4.70 K and never crossed it; 25 of 64 walkers did.

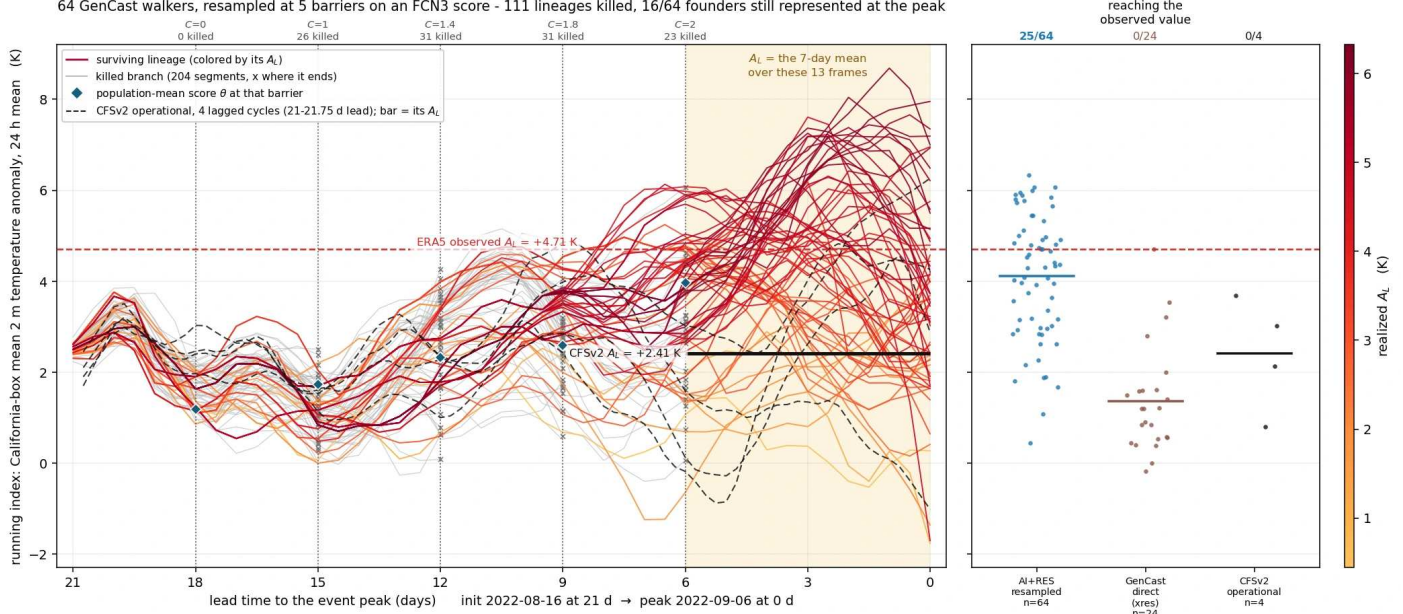
OBSERVED A_L +4.71 KERA5, 35-41 N, 124-118 W
CLIMATOLOGICAL PERCENTILE 99.71thERA5 1959-2022 · 99.84th against 1990-2019
RETURN PERIOD 37 yr2 of 64 (2017, 2022) years hold a window this extreme
DEPTH IN GENCAST'S SPREAD ~+2.9 σweek-3 ensemble not tabulated
DIRECT SAMPLING 0 / 24best member +4.70 K · P = 0.000 [0.000, 0.142]
AI+RES 25 / 64most extreme +6.33 K
WEIGHTED P(A_L BEYOND OBSERVED) 0.0186sum of the walker probabilities past the observation
PEAK-DAY ANOMALY +7.42 Kthe single day, not the week · 100.0th percentile
64-YEAR RECORD FOR THIS SEASON +6.30 K (2022-09-10)peak 2022-09-06 · init 2022-08-16 (21 d lead)

Sacramento's all-time 116 °F fell on this peak. The box is the named region, not the week's maximum - the unconstrained maximum for that week is +7.32 K over Idaho and Montana and was deliberately not used, because a box there would score a different event than the one it names. Calibration is clean where direct resolves: $P(\geq +2.99 \text{ K})$ 0.114 against 0.125, $P(\geq +3.99 \text{ K})$ 0.045 against 0.042, and the curve runs out to 7.7×10^{-5} at +6.33 K.



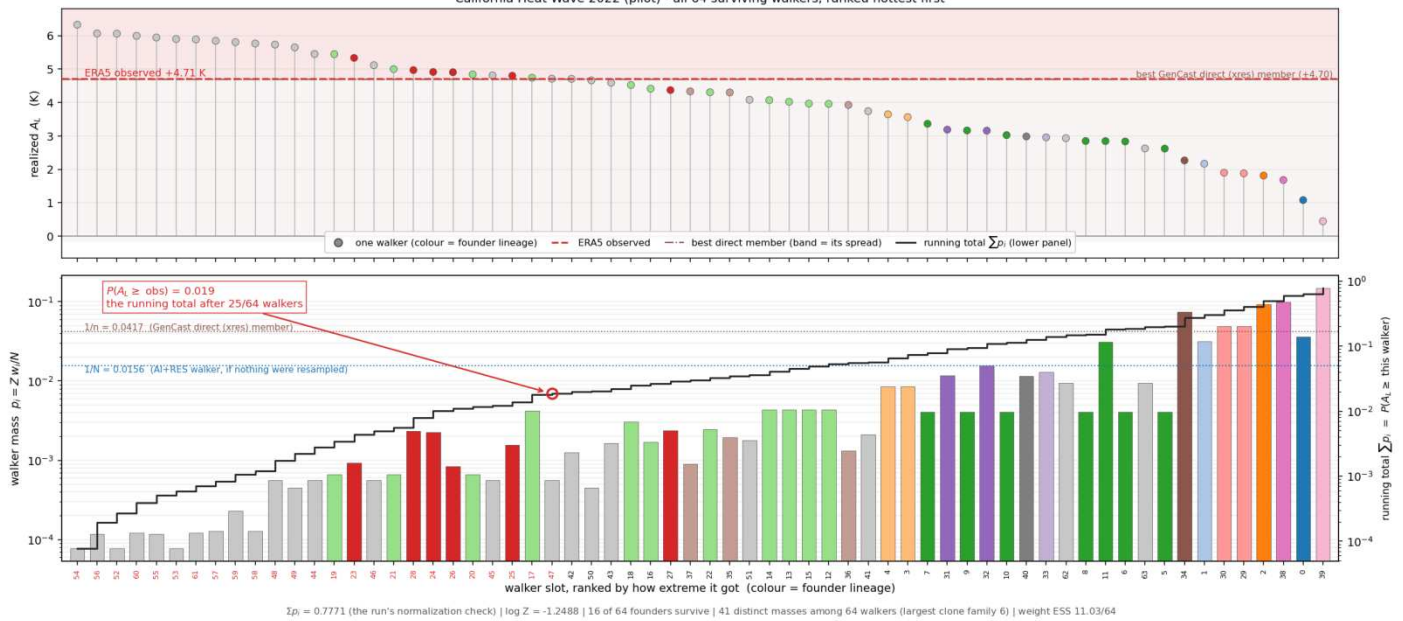
AI+RES pilot - California_HeatWave_2022 (pilot): the walk itself, barrier by barrier

64 GenCast walkers, resampled at 5 barriers on an FCN3 score - 111 lineages killed, 16/64 founders still represented at the peak



Trajectory. The algorithm happening: A_L against lead time for every branch the run rolled, grey for lineages that were killed and coloured for the 64 that survived, with the resampling barriers as vertical lines, the 7-day verification window shaded, the observed value in red, and the operational CFSv2 forecast at the same lead in black.

California Heat Wave 2022 (pilot) - all 64 surviving walkers, ranked hottest first



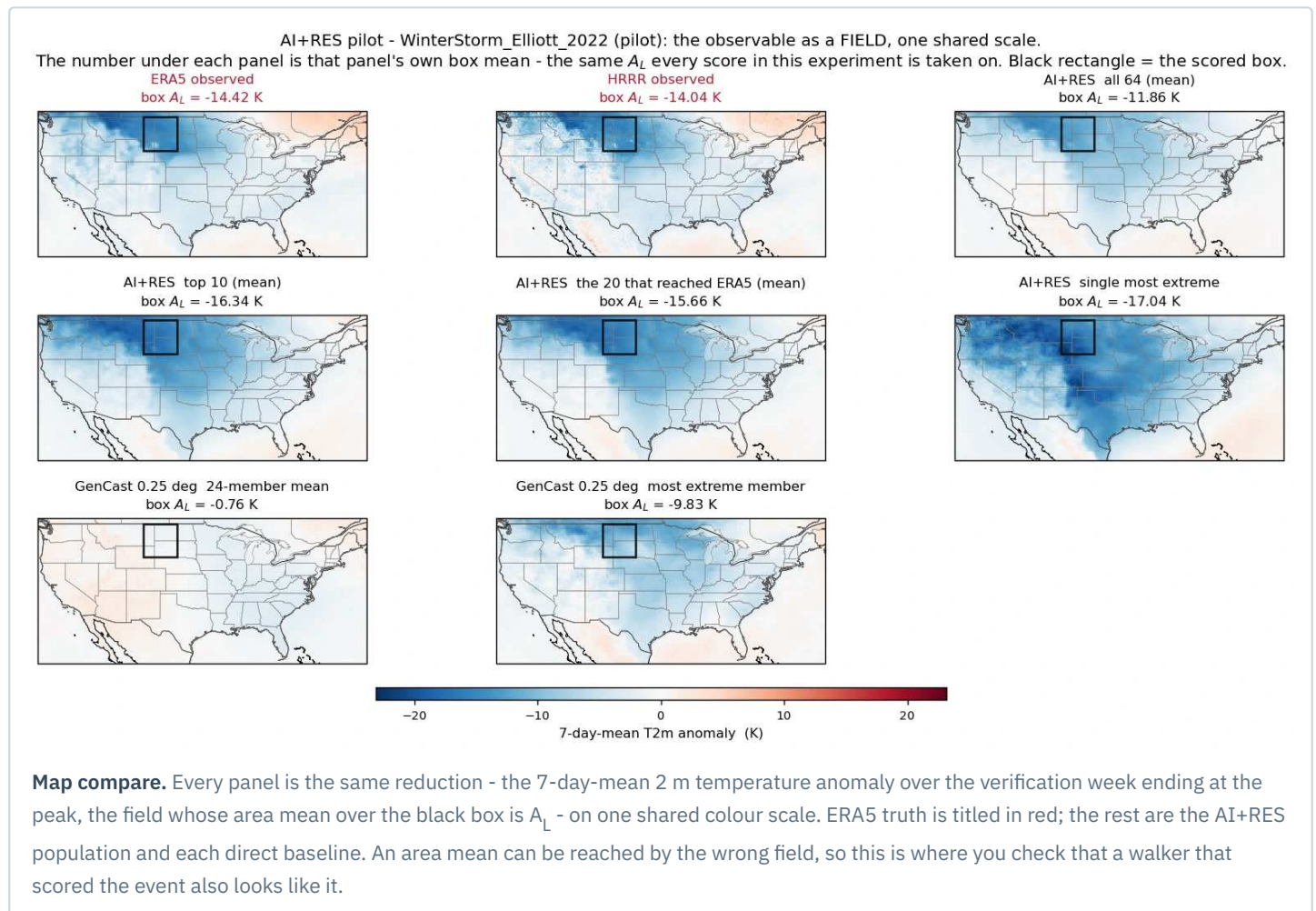
Every walker, and what it is worth. Top: all 64 surviving walkers ranked by how extreme they got, against the observed value in red and the best direct-sampling member in brown. Bottom: the probability mass $p_i = Z \cdot w_i / N$ each one carries, on a log axis, with the running total $\sum p_i$ as the black staircase - the headline probability is where that staircase stands when the ranking crosses the observation. Two reference lines set the scale: $1/n$, what a single direct member is worth, and $1/N$, what a walker would be worth if nothing had been resampled. Bars at identical height are clones of one parent at the last resampling and carry identical mass by construction, so the count of distinct masses under the panel - not 64 - is how many independent pieces of information the estimate rests on.

The deepest reach of the whole experiment, and the second cold hit. The observation sits 0.5σ deeper in the model's distribution than Southwest, which was missed - and this one is reached, overshooting by 2.6 K.

OBSERVED A_L -14.42 KERA5, 43-49 N, 106-100 W
CLIMATOLOGICAL PERCENTILE 98.41 th ERA5 1959-2022 · 99.13th against 1990-2019
RETURN PERIOD 5 yr11 of 64 years hold a window this extreme
DEPTH IN GENCAST'S SPREAD +2.81 σweek-3 ensemble -0.76 ± 4.87 K
DIRECT SAMPLING 0 / 24 best member -9.83 K · $P = 0.000$ [0.000, 0.142]
AI+RES 20 / 64 most extreme -17.04 K
WEIGHTED $P(A_L \text{ BEYOND OBSERVED})$ 0.0079 sum of the walker probabilities past the observation
PEAK-DAY ANOMALY -19.71 Kthe single day, not the week · 99.0th percentile
64-YEAR RECORD FOR THIS SEASON -21.35 K (1983-12-25)peak 2022-12-23 · init 2022-12-02 (21 d lead)

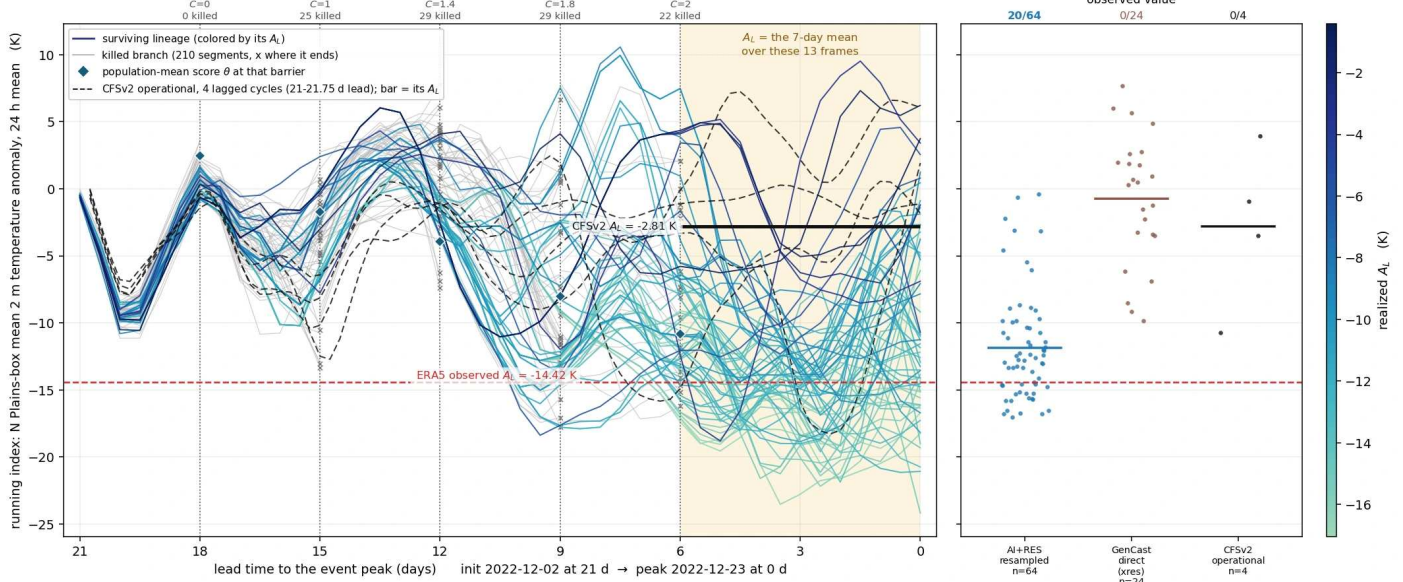
box: Northern Plains (43-49 N, 106-100 W) · registry key WinterStorm_Elliott_2022 · Slurm job 1184, 7 h 26 m
run health: log Z -1.1612 · normalization check 0.8108, the closest to 1 of the wave · 14/64
founders · ESS/N 1.00 / 0.39 / 0.29 / 0.33 / 0.59 · weight ESS 8.61

The free second rung comes at no extra GPU cost: the run records the CONUS index alongside the box index, giving a second calibration anchor on the cold side at -2.73 K, where direct sampling returns $1/24 = 0.042$ [0.007, 0.202] and AI+RES returns 0.136 - inside the interval. The tilt is only partly undone here (unweighted -11.86 K to weighted -3.64 K against a direct mean of -0.76 K), which is the pattern across the wave: the deeper the tilt, the less of it the reweighting recovers.

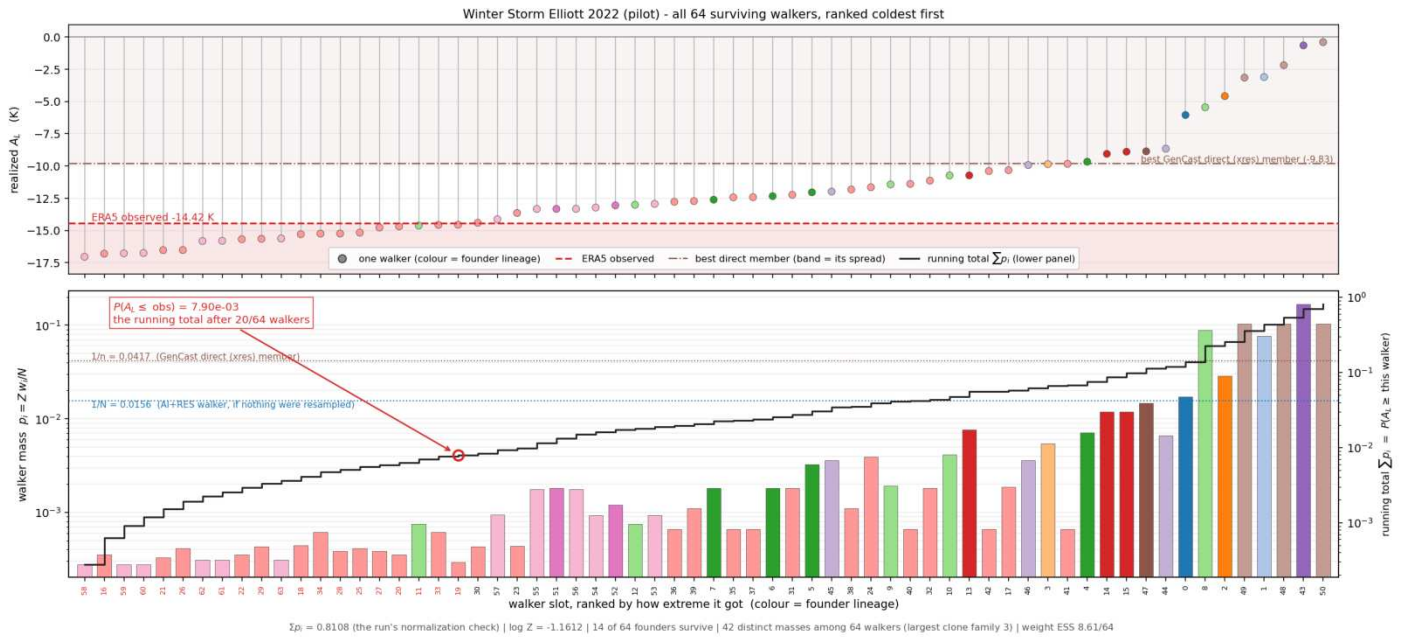


AI+RES pilot - WinterStorm_Elliott_2022 (pilot): the walk itself, barrier by barrier

64 GenCast walkers, resampled at 5 barriers on an FCN3 score - 105 lineages killed, 14/64 founders still represented at the peak



Trajectory. The algorithm happening: A_L against lead time for every branch the run rolled, grey for lineages that were killed and coloured for the 64 that survived, with the resampling barriers as vertical lines, the 7-day verification window shaded, the observed value in red, and the operational CFSv2 forecast at the same lead in black.



Every walker, and what it is worth. Top: all 64 surviving walkers ranked by how extreme they got, against the observed value in red and the best direct-sampling member in brown. Bottom: the probability mass $p_i = Z \cdot w_i / N$ each one carries, on a log axis, with the running total $\sum p_i$ as the black staircase - the headline probability is where that staircase stands when the ranking crosses the observation. Two reference lines set the scale: $1/n$, what a single direct member is worth, and $1/N$, what a walker would be worth if nothing had been resampled. Bars at identical height are clones of one parent at the last resampling and carry identical mass by construction, so the count of distinct masses under the panel - not 64 - is how many independent pieces of information the estimate rests on.

Winter Storm Uri

February 2021

RUNG 4 OF 9

cold reached

The first cold hit, and the first clean demonstration that the weights are honest. The frozen schedule, carried unchanged from four warm heat events to a Texas freeze, reaches the observation and prices it.

OBSERVED A_L

-7.40 K ERA5, 26-37 N, 106-94 W

CLIMATOLOGICAL PERCENTILE

96.80th ERA5 1959-2022 · 98.45th against 1990-2019

RETURN PERIOD

4 yr 19 of 64 years hold a window this extreme

DEPTH IN GENCAST'S SPREAD

+2.28 σ week-3 ensemble $+0.06 \pm 3.27$ K

DIRECT SAMPLING

0 / 24 best member -6.51 K · P = 0.000 [0.000, 0.142]

AI+RES

32 / 64 most extreme -10.20 K

WEIGHTED P(A_L BEYOND OBSERVED)

0.0090 sum of the walker probabilities past the observation

PEAK-DAY ANOMALY

-19.95 K the single day, not the week · 100.0th percentile

64-YEAR RECORD FOR THIS SEASON

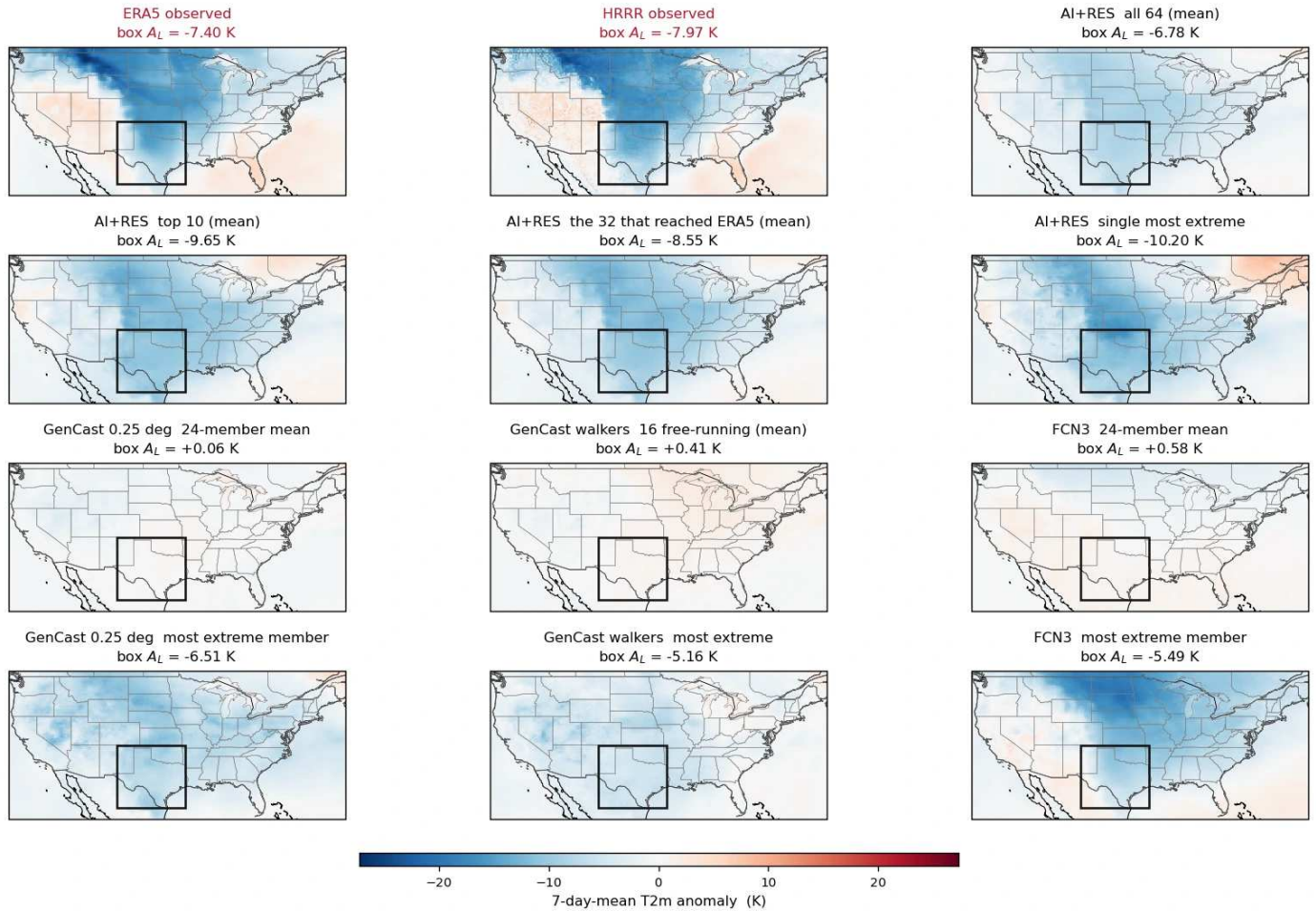
-14.73 K (2021-02-20) peak 2021-02-15 · init 2021-01-25 (21 d lead)

box: Texas and Oklahoma (26-37 N, 106-94 W) · registry key WinterStorm Uri_2021 · Slurm job 1183, 7 h 12 m
run health: log Z -0.6735 · normalization check 1.7688 · 16/64 founders · ESS/N 1.00 / 0.40 / 0.43 / 0.41 / 0.30 · weight ESS only 2.24 of 64, so read the spatial map as shape, not level

Three independent direct baselines are empty here - 16 free-running walkers, the 24-member GenCast cube and the 24-member FCN3 ensemble - with the best member 0.9 K short. The tilt is fully undone: an unweighted mean of -6.78 K reweights to +0.21 K against a direct mean of +0.41 K, and the population still covers the bulk (its range reaches +1.57 K). Curve agreement is the best of the wave, leaning slightly low if anything, then extending about two decades past where every direct curve stops. One number here needs reading carefully: a 4-year return period looks mild for the storm that took down the Texas grid, and it is an artefact of the observable, not of Uri. A 6-day box mean over 26-37 N smooths a 3-day collapse across eleven degrees of latitude; the single peak day was -19.95 K, the most extreme mid-February day in the whole 64-year record for that box. What made Uri catastrophic was duration and minimum temperature in a place with no winterisation, and A_L is not built to see either.

AI+RES pilot - WinterStorm_Uri_2021 (pilot): the observable as a FIELD, one shared scale.

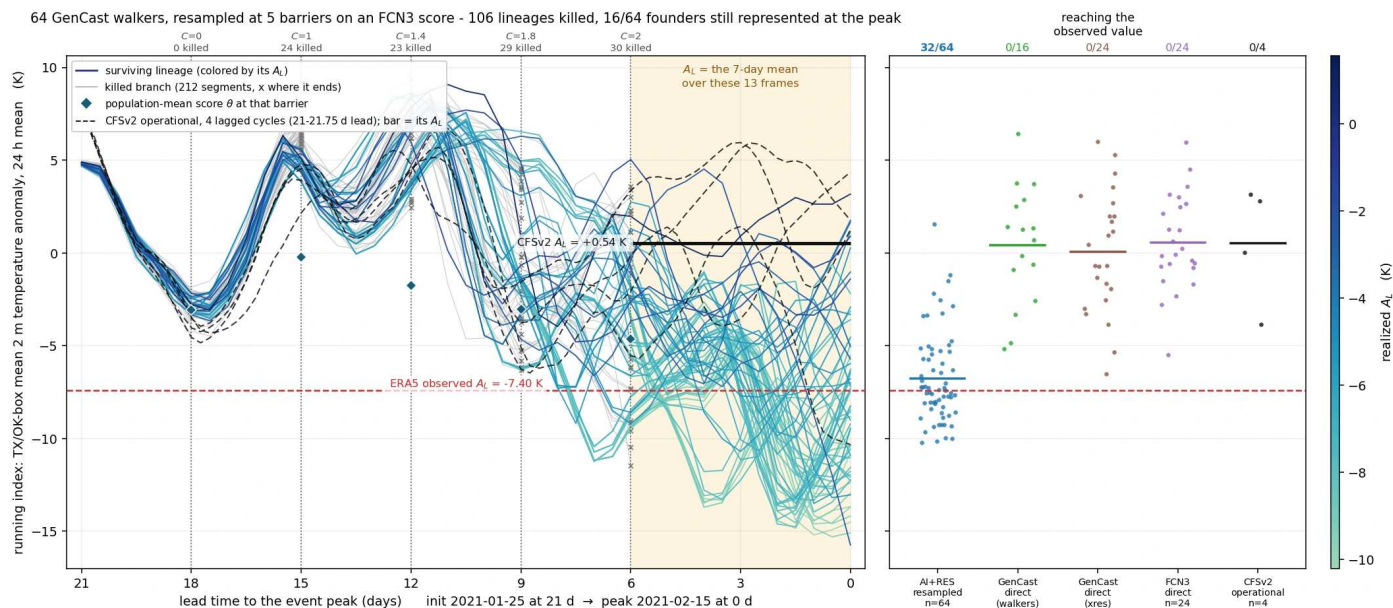
The number under each panel is that panel's own box mean - the same A_L every score in this experiment is taken on. Black rectangle = the scored box.



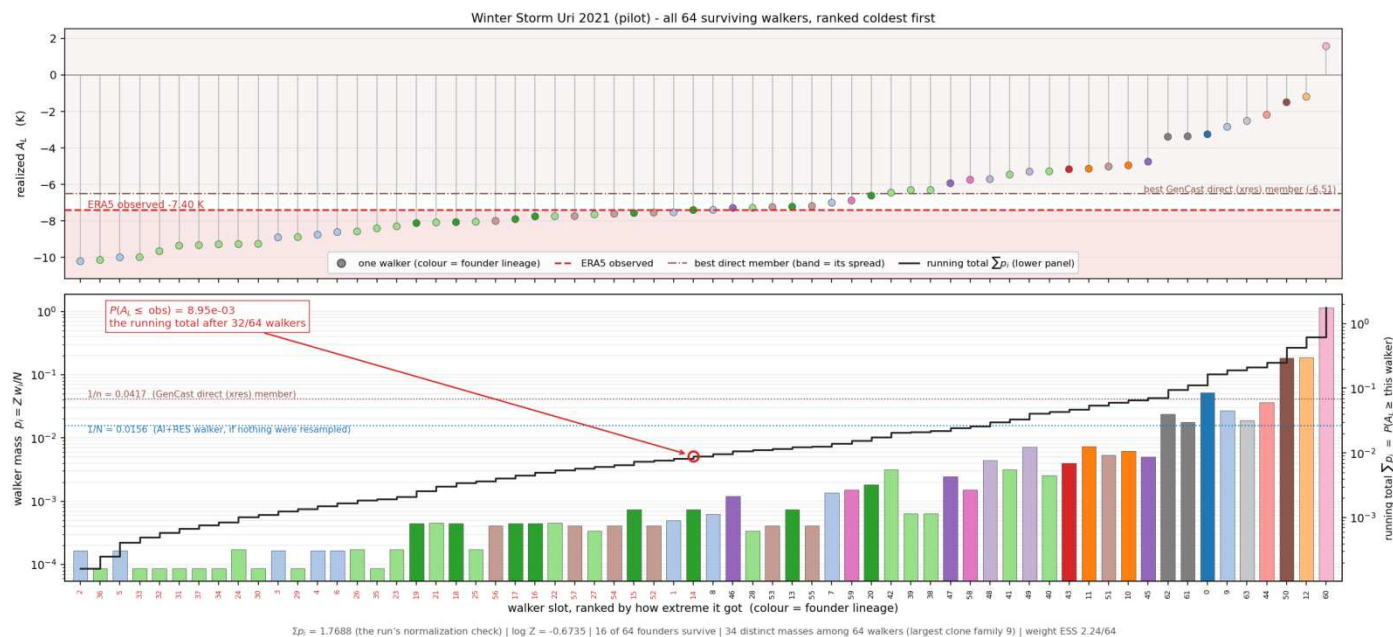
Map compare. Every panel is the same reduction - the 7-day-mean 2 m temperature anomaly over the verification week ending at the peak, the field whose area mean over the black box is A_L - on one shared colour scale. ERA5 truth is titled in red; the rest are the AI+RES population and each direct baseline. An area mean can be reached by the wrong field, so this is where you check that a walker that scored the event also looks like it.

AI+RES pilot - WinterStorm_Uri_2021 (pilot): the walk itself, barrier by barrier

64 GenCast walkers, resampled at 5 barriers on an FCN3 score - 106 lineages killed, 16/64 founders still represented at the peak



Trajectory. The algorithm happening: A_L against lead time for every branch the run rolled, grey for lineages that were killed and coloured for the 64 that survived, with the resampling barriers as vertical lines, the 7-day verification window shaded, the observed value in red, and the operational CFSv2 forecast at the same lead in black.

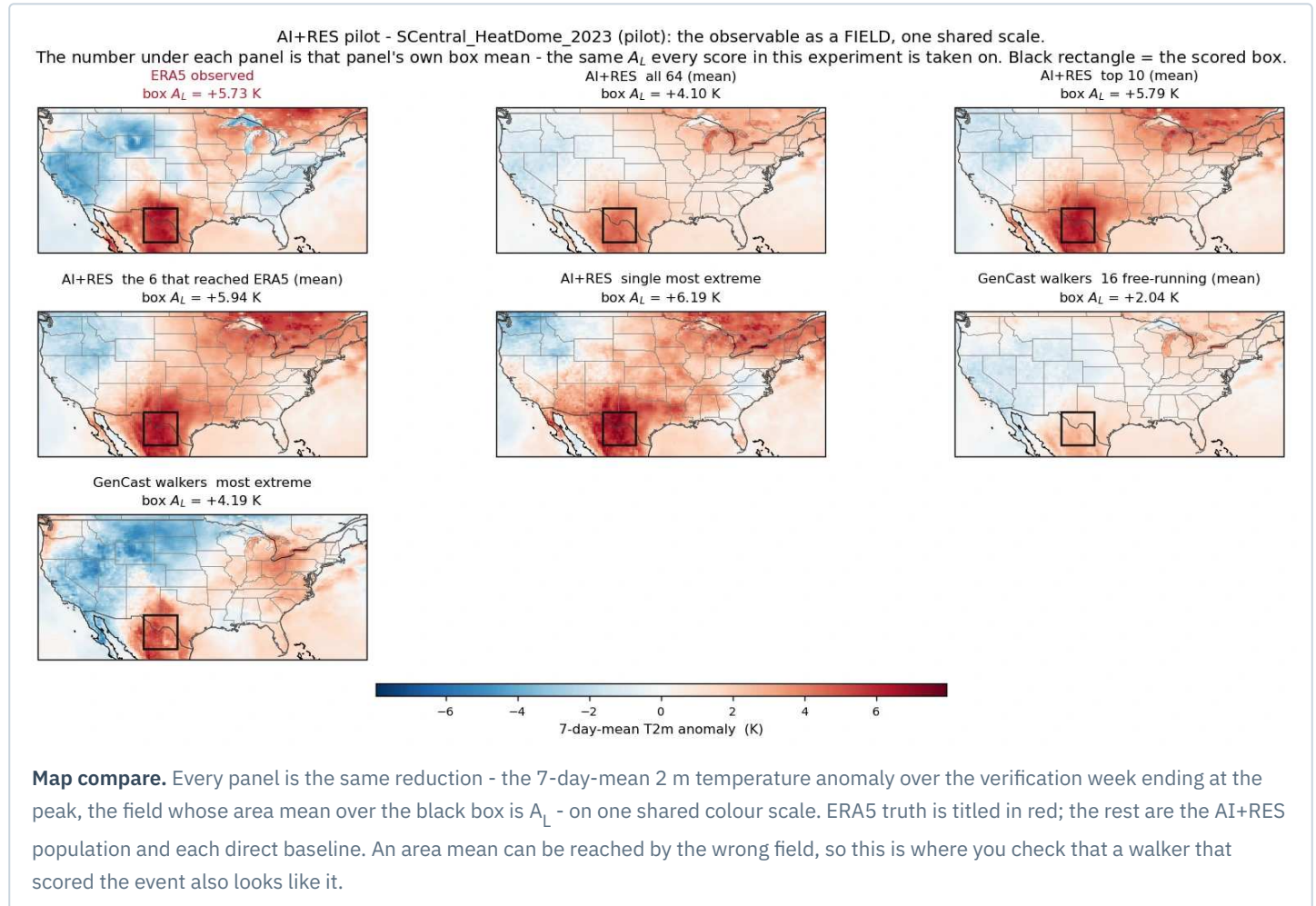


Every walker, and what it is worth. Top: all 64 surviving walkers ranked by how extreme they got, against the observed value in red and the best direct-sampling member in brown. Bottom: the probability mass $p_i = Z \cdot w_i / N$ each one carries, on a log axis, with the running total $\sum p_i$ as the black staircase - the headline probability is where that staircase stands when the ranking crosses the observation. Two reference lines set the scale: $1/n$, what a single direct member is worth, and $1/N$, what a walker would be worth if nothing had been resampled. Bars at identical height are clones of one parent at the last resampling and carry identical mass by construction, so the count of distinct masses under the panel - not 64 - is how many independent pieces of information the estimate rests on.

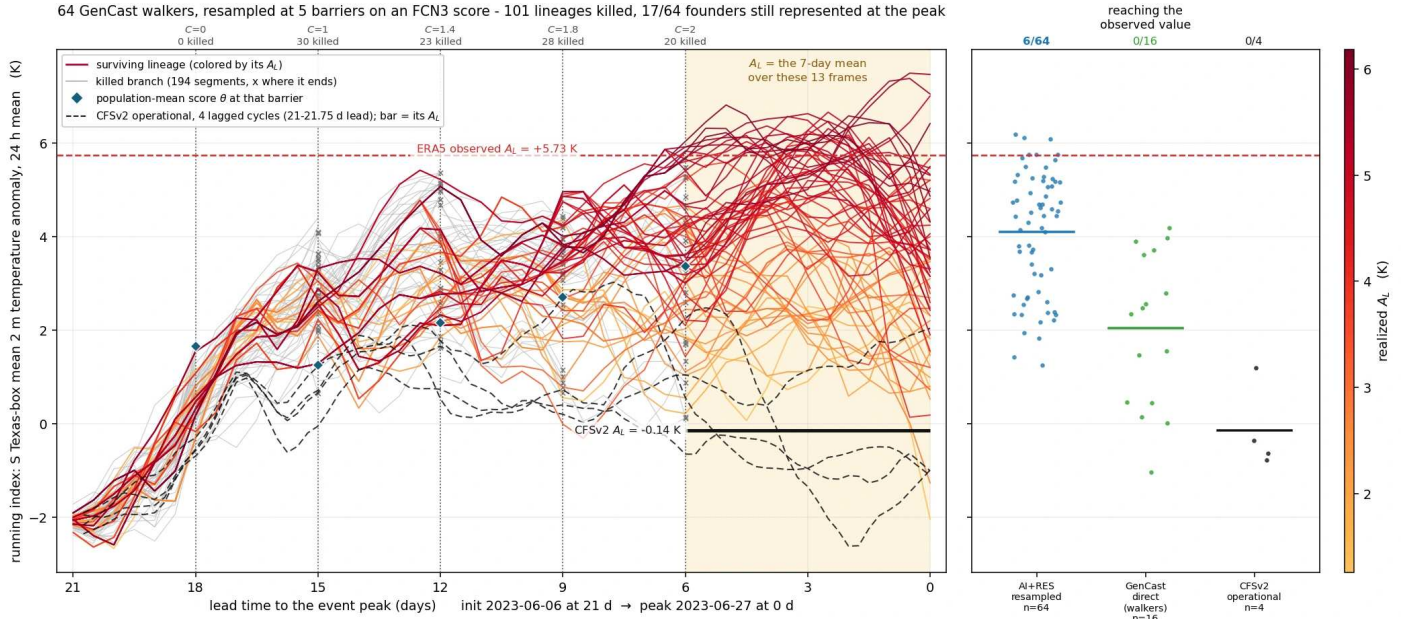
The rarest event on the slate against its own climatology: no window in 64 years of late-June Texas comes within 0.75 K of it. The frozen schedule, ported unchanged into a subtropical-ridge regime, reaches it.

OBSERVED A_L +5.73 K _{ERA5, 26-32 N, 106-100 W}
CLIMATOLOGICAL PERCENTILE 100.00 _{th ERA5 1959-2022 · 100.00th against 1990-2019}
RETURN PERIOD beyond the record _{0 of 64 years hold a window this extreme}
DEPTH IN GENCAST'S SPREAD ~+2.2 σ _{week-3 ensemble not tabulated}
DIRECT SAMPLING 0 / 16 _{best member +4.19 K · P = 0.000 [0.000, 0.206]}
AI+RES 6 / 64 _{most extreme +6.19 K}
WEIGHTED P(A_L BEYOND OBSERVED) 0.00273 _{sum of the walker probabilities past the observation}
PEAK-DAY ANOMALY +5.54 K _{the single day, not the week · 100.0th percentile}
64-YEAR RECORD FOR THIS SEASON +4.98 K (2011-06-21) _{peak 2023-06-27 · init 2023-06-06 (21 d lead)}

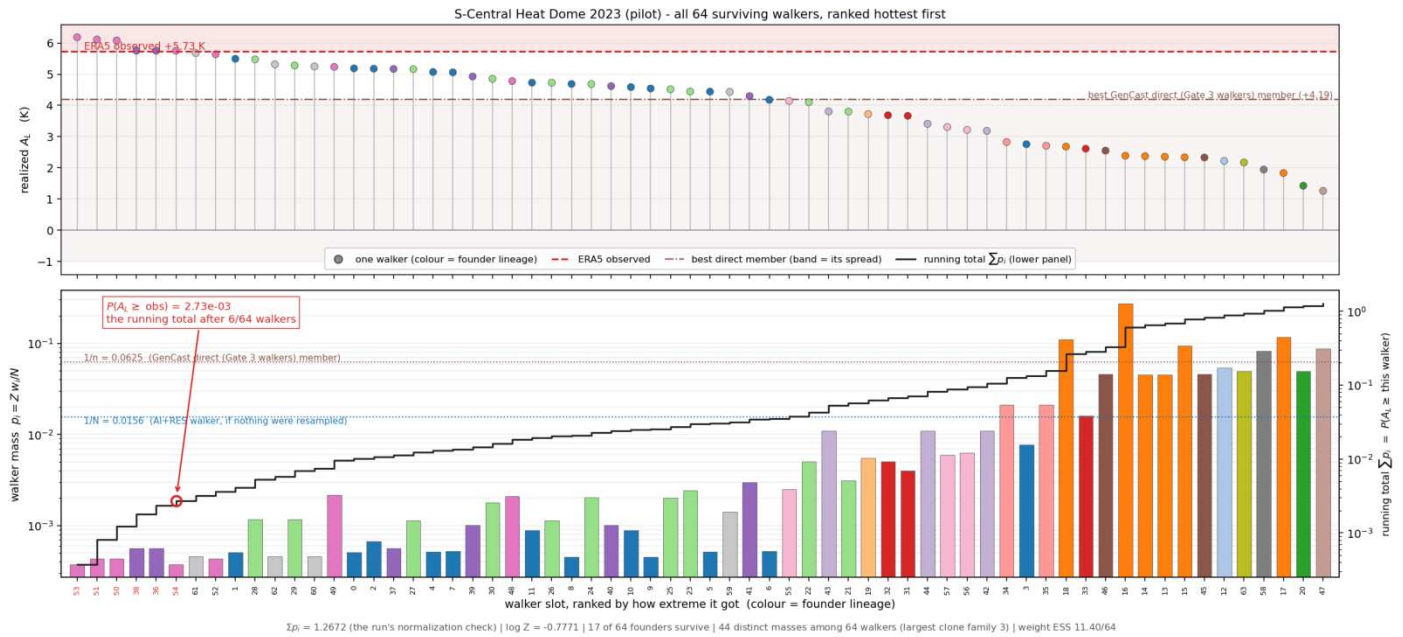
This is the generality claim. The C schedule was tuned once, on the Pacific Northwest, and has not been re-tuned for any event since - so a reach in a different regime is evidence the schedule generalises rather than that it was fitted. The weights stay honest: weighted mean +2.35 K against a direct mean of +2.04 K. The dome's core extended well into Mexico, south of the 24 N crop, so like the Pacific Northwest box this one sees the northern part of the event.



AI+RES pilot - S-Central_HeatDome_2023 (pilot): the walk itself, barrier by barrier



Trajectory. The algorithm happening: A_L against lead time for every branch the run rolled, grey for lineages that were killed and coloured for the 64 that survived, with the resampling barriers as vertical lines, the 7-day verification window shaded, the observed value in red, and the operational CFSv2 forecast at the same lead in black.



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Pacific Northwest Heat Dome

June 2021

RUNG 6 OF 9

heat reached

The pilot, and the run every other one is measured against. Forty direct GenCast members initialised three weeks out never reach the dome; 42 of 64 walkers do.

OBSERVED A_L

+7.72 K ERA5, 44-50 N, 124-118 W

CLIMATOLOGICAL PERCENTILE

99.60 ERA5 1959-2022 · 99.61th against 1990-2019

RETURN PERIOD

22 yr 3 of 64 (1992, 2015, 2021) years hold a window this extreme

DEPTH IN GENCAST'S SPREAD

+1.98 σ week-3 ensemble $+2.46 \pm 2.65$ K

DIRECT SAMPLING

0 / 40 best member +7.49 K · $P = 0.000$ [0.000, 0.088]

AI+RES

42 / 64 most extreme +10.29 K

WEIGHTED $P(A_L \text{ BEYOND OBSERVED})$

0.054 sum of the walker probabilities past the observation

PEAK-DAY ANOMALY

+14.38 K the single day, not the week · 100.0th percentile

64-YEAR RECORD FOR THIS SEASON

+11.11 K - the same dome, 3 d later peak 2021-06-28 · init 2021-06-07 (21 d lead)

box: Oregon and Washington, the dome core (44-50 N, 124-118 W) · registry key PNW_HeatDome_2021

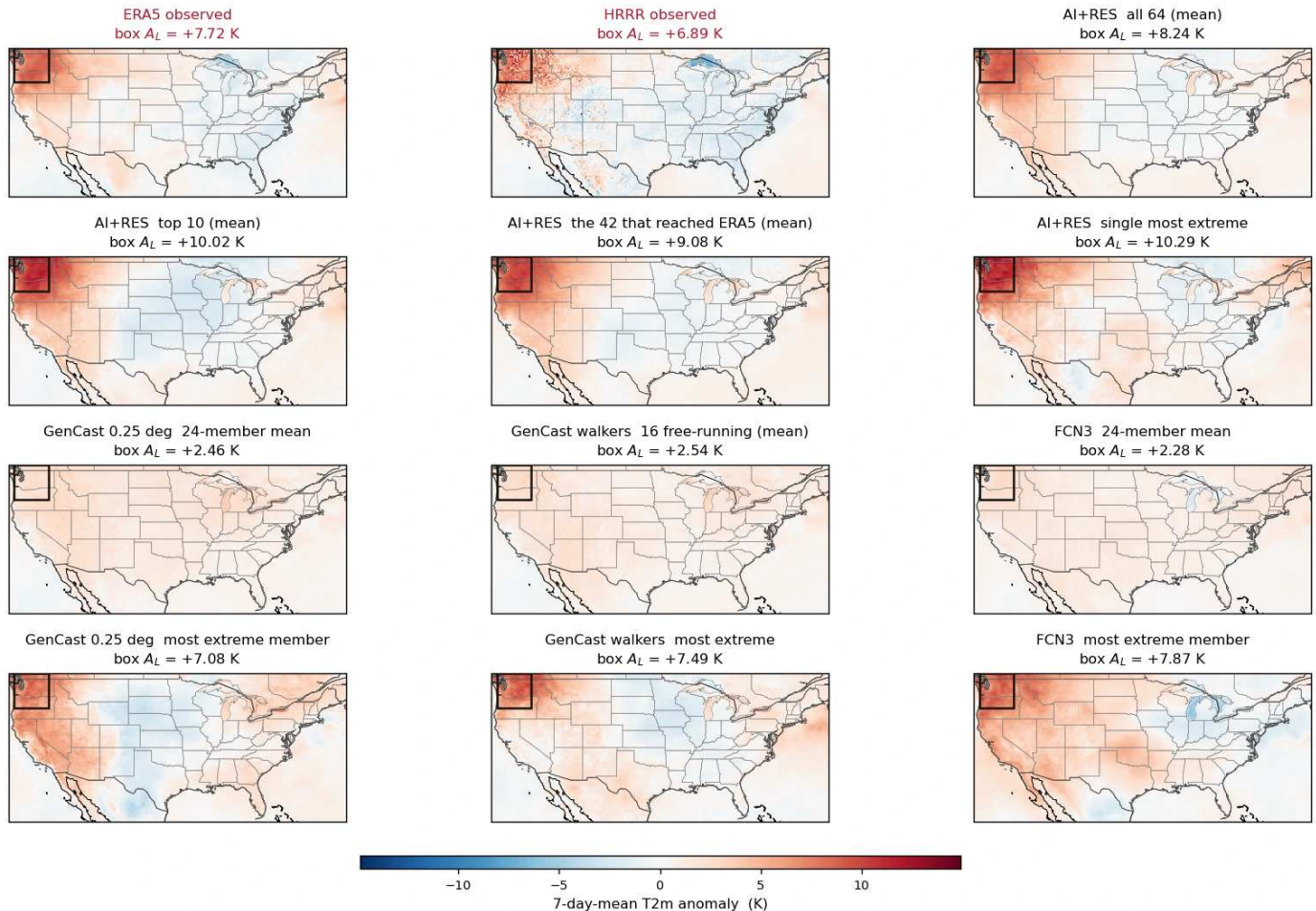
· Slurm job 1172, 7 h 45 m

run health: log Z -0.9317 · normalization check 0.5684 · 11/64 founders · weight ESS 5.68

The observed maximum reached about 52 N into British Columbia, outside the 50 N CONUS crop, so this box sees the southern two thirds of the event. The per-grid-point tail is where the estimate is most striking: the weighted spatial distribution puts $P(\text{point} \geq +9 \text{ K})$ at 0.0505 against ERA5's own 0.0848, where 24 direct members give 0.0011.

AI+RES pilot - PNW_HeatDome_2021 (pilot): the observable as a FIELD, one shared scale.

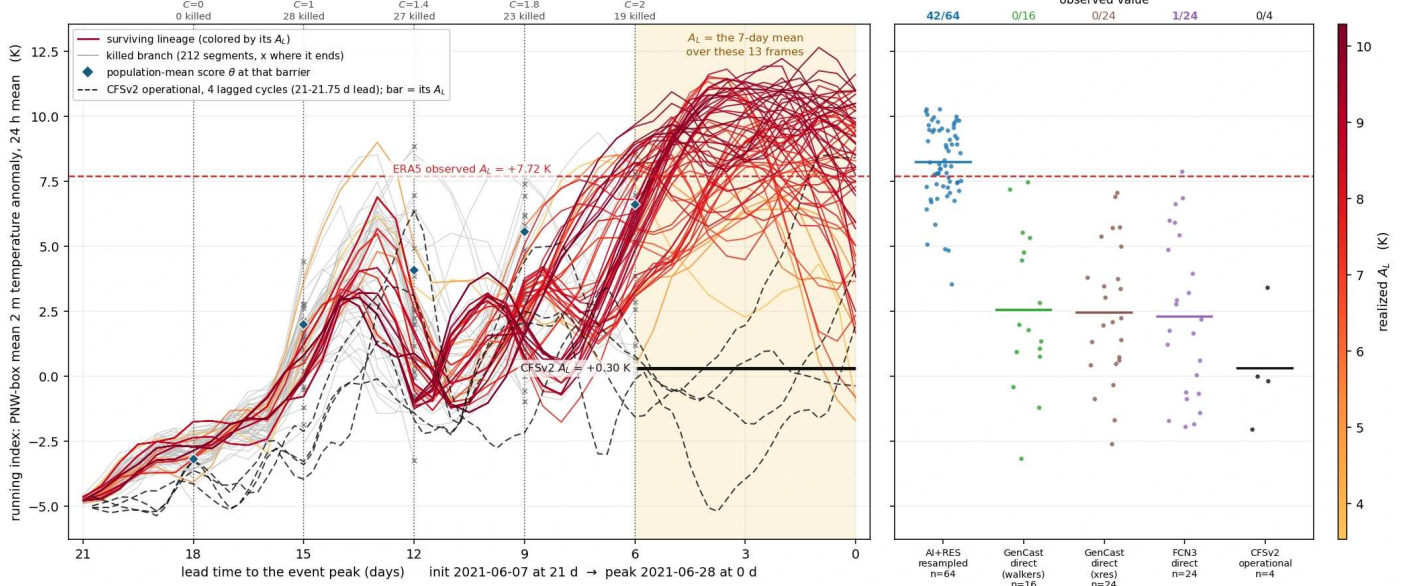
The number under each panel is that panel's own box mean - the same A_L every score in this experiment is taken on. Black rectangle = the scored box.



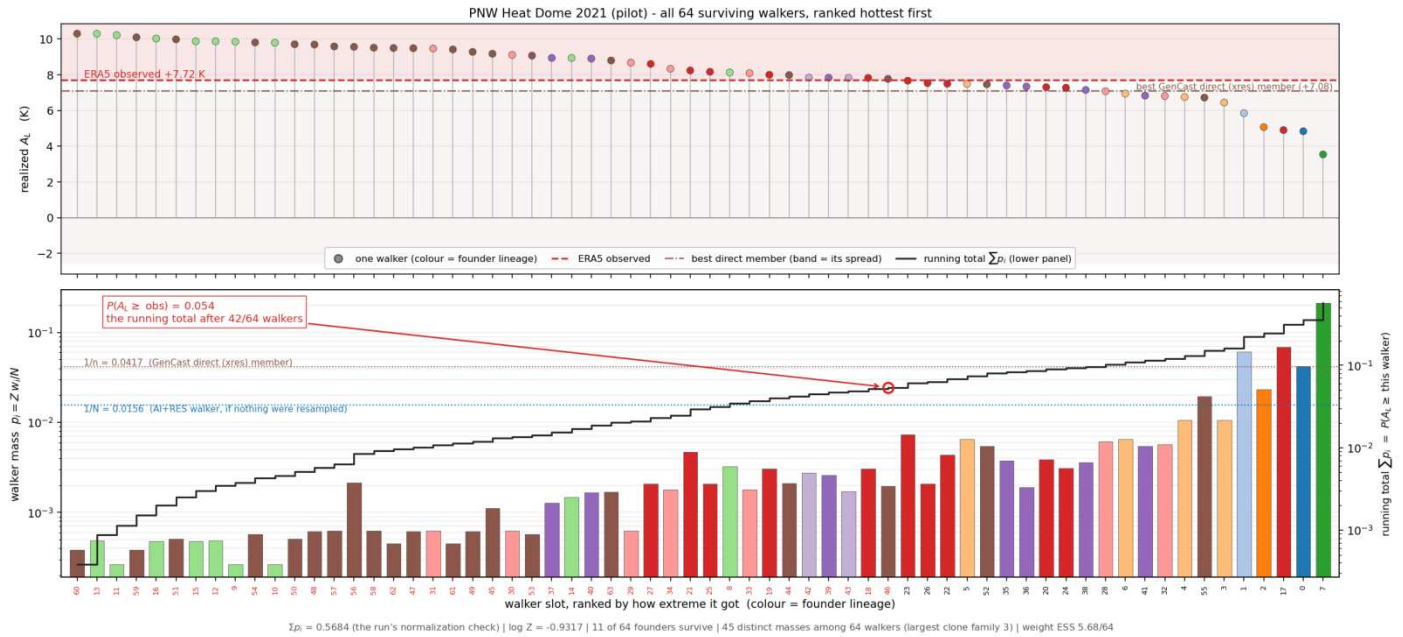
Map compare. Every panel is the same reduction - the 7-day-mean 2 m temperature anomaly over the verification week ending at the peak, the field whose area mean over the black box is A_L - on one shared colour scale. ERA5 truth is titled in red; the rest are the AI+RES population and each direct baseline. An area mean can be reached by the wrong field, so this is where you check that a walker that scored the event also looks like it.

AI+RES pilot - PNW_HeatDome_2021 (pilot): the walk itself, barrier by barrier

64 GenCast walkers, resampled at 5 barriers on an FCN3 score - 97 lineages killed, 11/64 founders still represented at the peak



Trajectory. The algorithm happening: A_L against lead time for every branch the run rolled, grey for lineages that were killed and coloured for the 64 that survived, with the resampling barriers as vertical lines, the 7-day verification window shaded, the observed value in red, and the operational CFSv2 forecast at the same lead in black.



Every walker, and what it is worth. Top: all 64 surviving walkers ranked by how extreme they got, against the observed value in red and the best direct-sampling member in brown. Bottom: the probability mass $p_i = Z \cdot w_i / N$ each one carries, on a log axis, with the running total $\sum p_i$ as the black staircase - the headline probability is where that staircase stands when the ranking crosses the observation. Two reference lines set the scale: $1/n$, what a single direct member is worth, and $1/N$, what a walker would be worth if nothing had been resampled. Bars at identical height are clones of one parent at the last resampling and carry identical mass by construction, so the count of distinct masses under the panel - not 64 - is how many independent pieces of information the estimate rests on.

The Nationwide Warm Spell

CONUS index

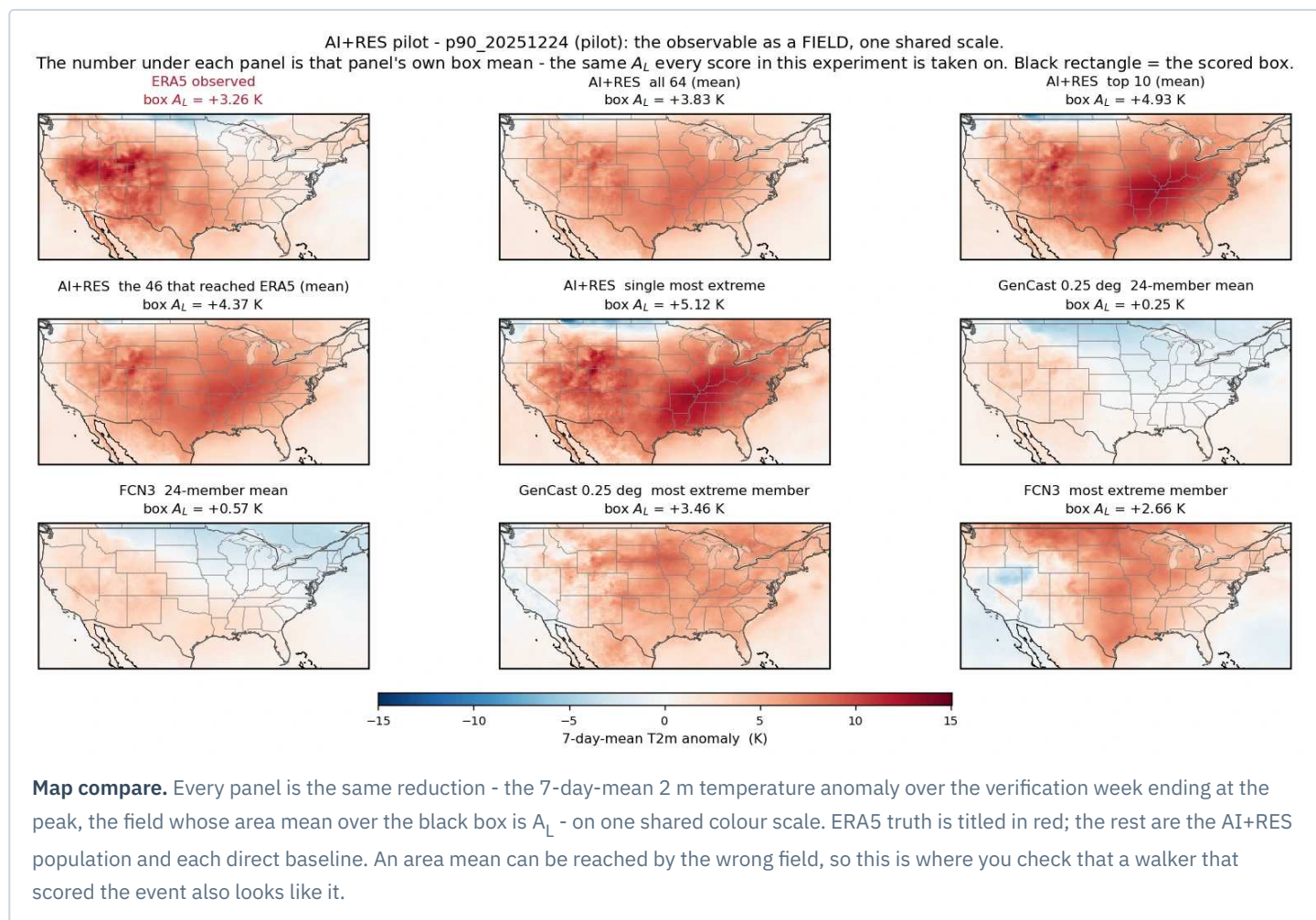
calibration anchor

The calibration anchor: the one rung where direct sampling returns a non-zero probability, so the weighted estimate has something to be checked against. It lands inside the direct interval.

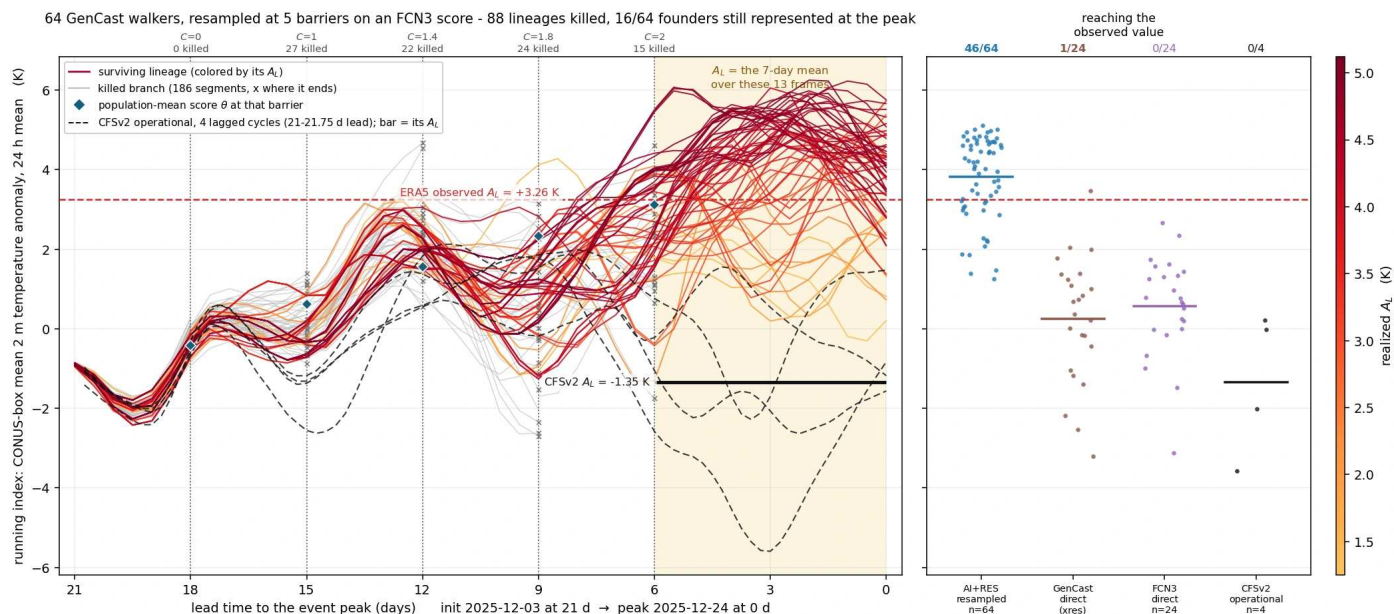
OBSERVED A_L +3.26 KERA5, 24-50 N, 125-66 W
CLIMATOLOGICAL PERCENTILE 98.78thERA5 1959-2022 · 97.40th against 1990-2019
RETURN PERIOD 10 yr8 of 64 years hold a window this extreme
DEPTH IN GENCAST'S SPREAD +1.91 σweek-3 ensemble +0.25 ± 1.58 K
DIRECT SAMPLING 1 / 24best member +3.46 K · P = 0.042 [0.001, 0.211]
AI+RES 46 / 64most extreme +5.12 K
WEIGHTED $P(A_L \text{ BEYOND OBSERVED})$ 0.0661sum of the walker probabilities past the observation
PEAK-DAY ANOMALY +4.91 Kthe single day, not the week · 99.8th percentile
64-YEAR RECORD FOR THIS SEASON +4.45 K (2015-12-15)peak 2025-12-24 · init 2025-12-03 (21 d lead)

box: Whole CONUS - this case is defined on the national mean (24-50 N, 125-66 W) · registry key p90_20251224 · Slurm job 1182, 8 h 42 m
run health: log Z -1.1846 · normalization check 0.6890 · 16/64 founders · ESS/N 1.00 / 0.32 / 0.56 / 0.54 / 0.69 - the only run that recovers after every barrier

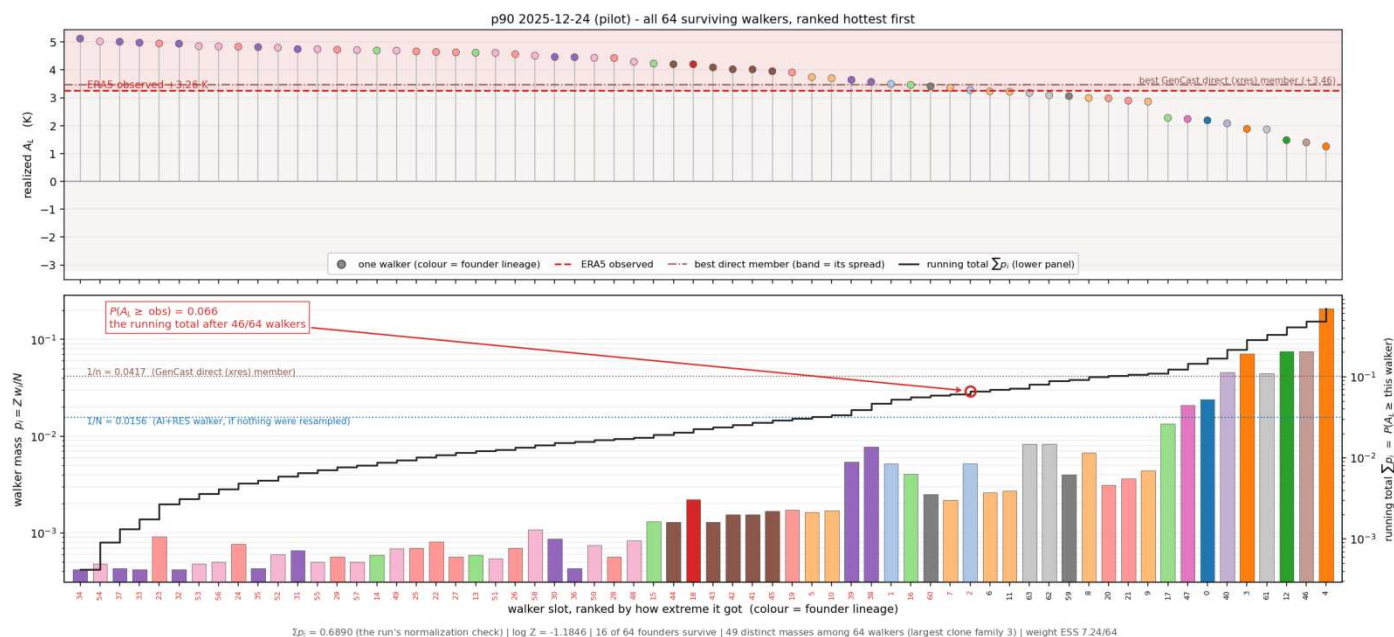
AI+RES gives 0.0661 against direct sampling's $1/24 = 0.0417$ [0.0074, 0.202]. No wave-1 run could do this - every one of them had direct at 0/24, so the weighted probability had nothing to be compared with. Two caveats: through the mid-range the weighted curve runs about twice the direct estimate (2.2 to 2.6 binomial standard errors at $n = 24$), and the weighted mean cannot return to the direct mean here, because resampling killed every lineage below +1.25 K. That is a limit of the surviving population's support, not a weighting error.



AI+RES pilot - p90_20251224 (pilot): the walk itself, barrier by barrier



Trajectory. The algorithm happening: A_L against lead time for every branch the run rolled, grey for lineages that were killed and coloured for the 64 that survived, with the resampling barriers as vertical lines, the 7-day verification window shaded, the observed value in red, and the operational CFSv2 forecast at the same lead in black.



Every walker, and what it is worth. Top: all 64 surviving walkers ranked by how extreme they got, against the observed value in red and the best direct-sampling member in brown. Bottom: the probability mass $p_i = Z \cdot w_i / N$ each one carries, on a log axis, with the running total $\sum p_i$ as the black staircase - the headline probability is where that staircase stands when the ranking crosses the observation. Two reference lines set the scale: $1/n$, what a single direct member is worth, and $1/N$, what a walker would be worth if nothing had been resampled. Bars at identical height are clones of one parent at the last resampling and carry identical mass by construction, so the count of distinct masses under the panel - not 64 - is how many independent pieces of information the estimate rests on.

The Normal-Day Test

2 August 2024

RUNG 8 OF 9

CONUS index

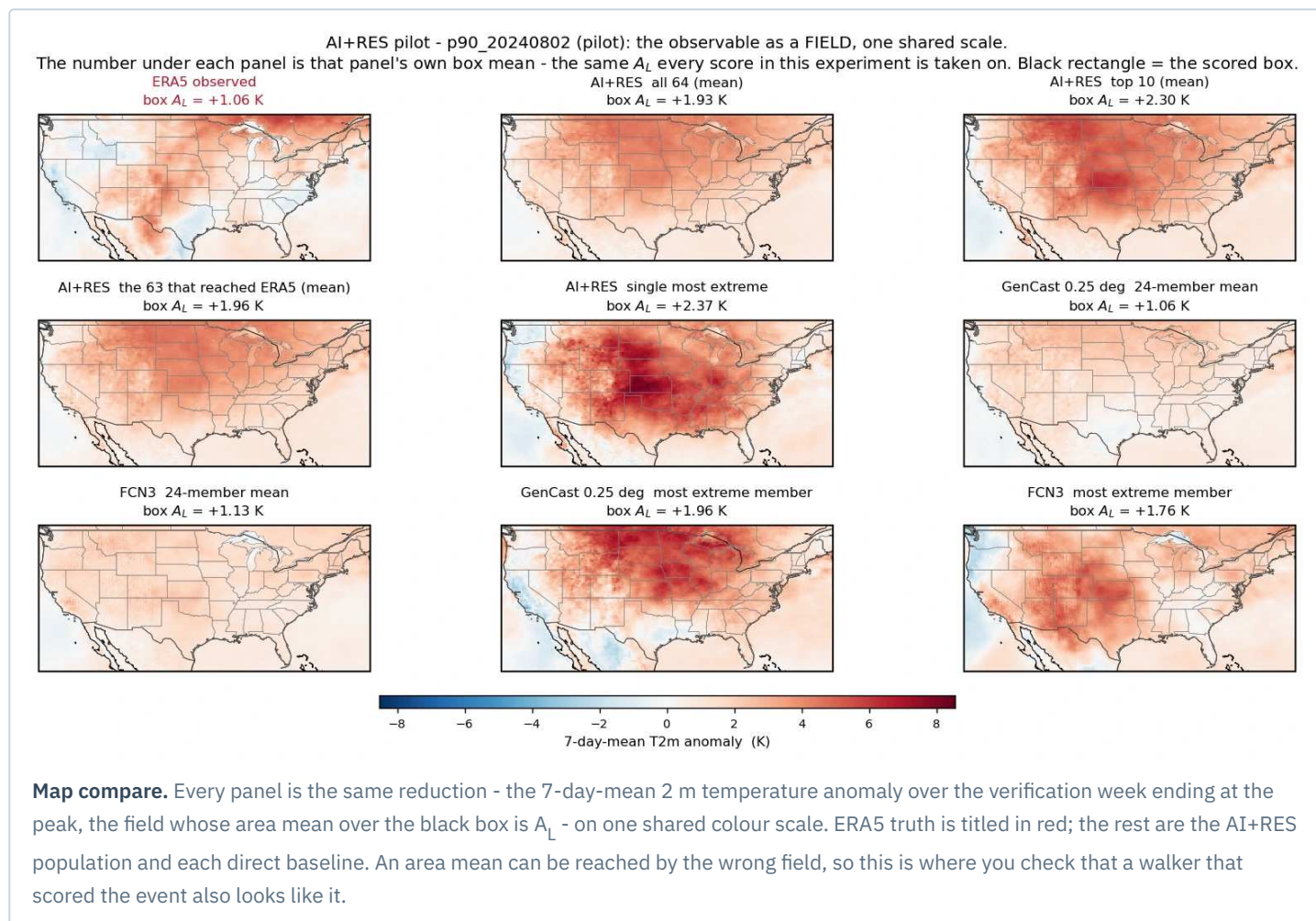
no-harm control

The no-harm control. The observation sits exactly at the ensemble median, and a sampler built to push walkers into the tail has to return an ordinary probability for an ordinary day. It returns 0.649 against direct sampling's 0.583.

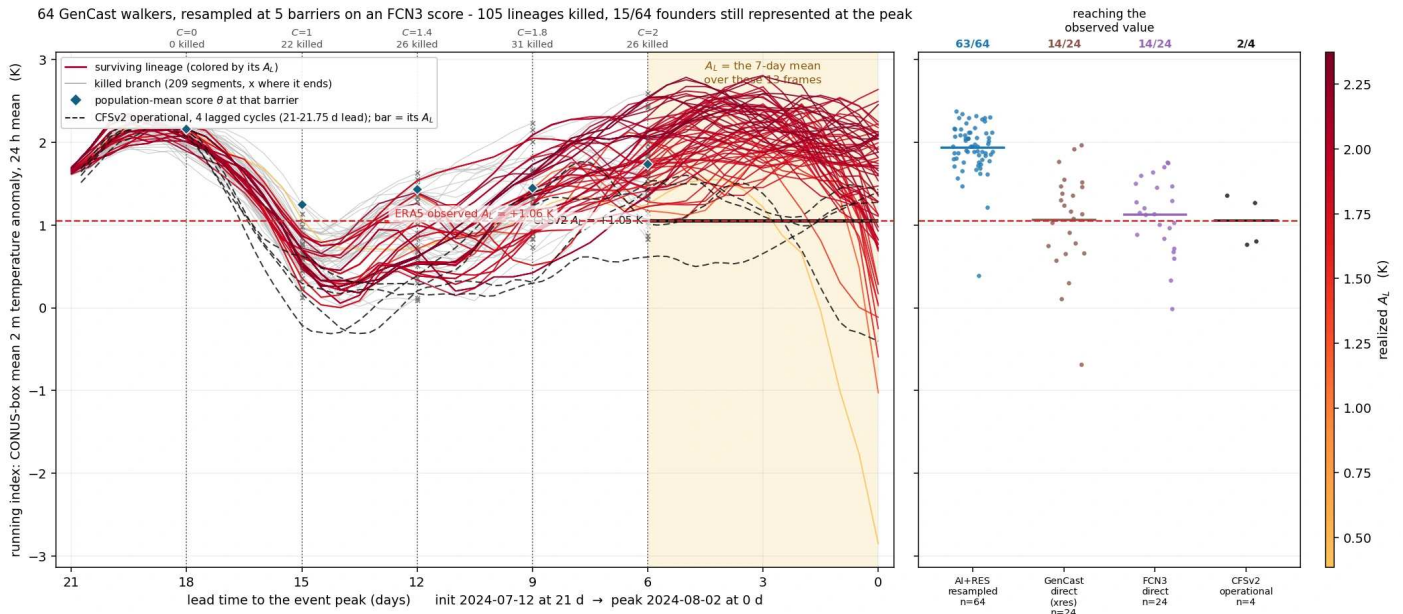
OBSERVED A_L +1.06 K ERA5, 24-50 N, 125-66 W
CLIMATOLOGICAL PERCENTILE 94.08th ERA5 1959-2022 · 91.55th against 1990-2019
RETURN PERIOD 4 yr 20 of 64 years hold a window this extreme
DEPTH IN GENCAST'S SPREAD 0.00 σ week-3 ensemble +1.06 ± 0.60 K
DIRECT SAMPLING 14 / 24 best member +1.96 K · P = 0.583 [0.366, 0.779]
AI+RES 63 / 64 most extreme +2.37 K
WEIGHTED P(A_L BEYOND OBSERVED) 0.649 sum of the walker probabilities past the observation
PEAK-DAY ANOMALY +2.40 K the single day, not the week · 100.0th percentile
64-YEAR RECORD FOR THIS SEASON +2.06 K (2006-07-20) peak 2024-08-02 · init 2024-07-12 (21 d lead)

box: Whole CONUS - this case is defined on the national mean (24-50 N, 125-66 W) · registry key p90_20240802 · Slurm job 1185, 7 h 48 m
run health: log Z -0.6063 · normalization check 0.7039 · 15/64 founders · ESS/N 1.00 / 0.53 / 0.37 / 0.30 / 0.40

Do not read the reach count as a result: 63 of 64 walkers 'reach' a median target, which says where the tilt pushed the population and nothing else. What matters is that the estimator was built to detect a specific failure - reporting 0.05 for a median target - and does not show it. The spatial evidence is cleaner still: on the per-grid-point tail the weighted population gives 0.0727 against direct GenCast's 0.0702, agreement to 4 %, while the unweighted population gives 0.1627, 2.3 times too heavy.



AI+RES pilot - p90_20240802 (pilot): the walk itself, barrier by barrier



Trajectory. The algorithm happening: A_L against lead time for every branch the run rolled, grey for lineages that were killed and coloured for the 64 that survived, with the resampling barriers as vertical lines, the 7-day verification window shaded, the observed value in red, and the operational CFSv2 forecast at the same lead in black.



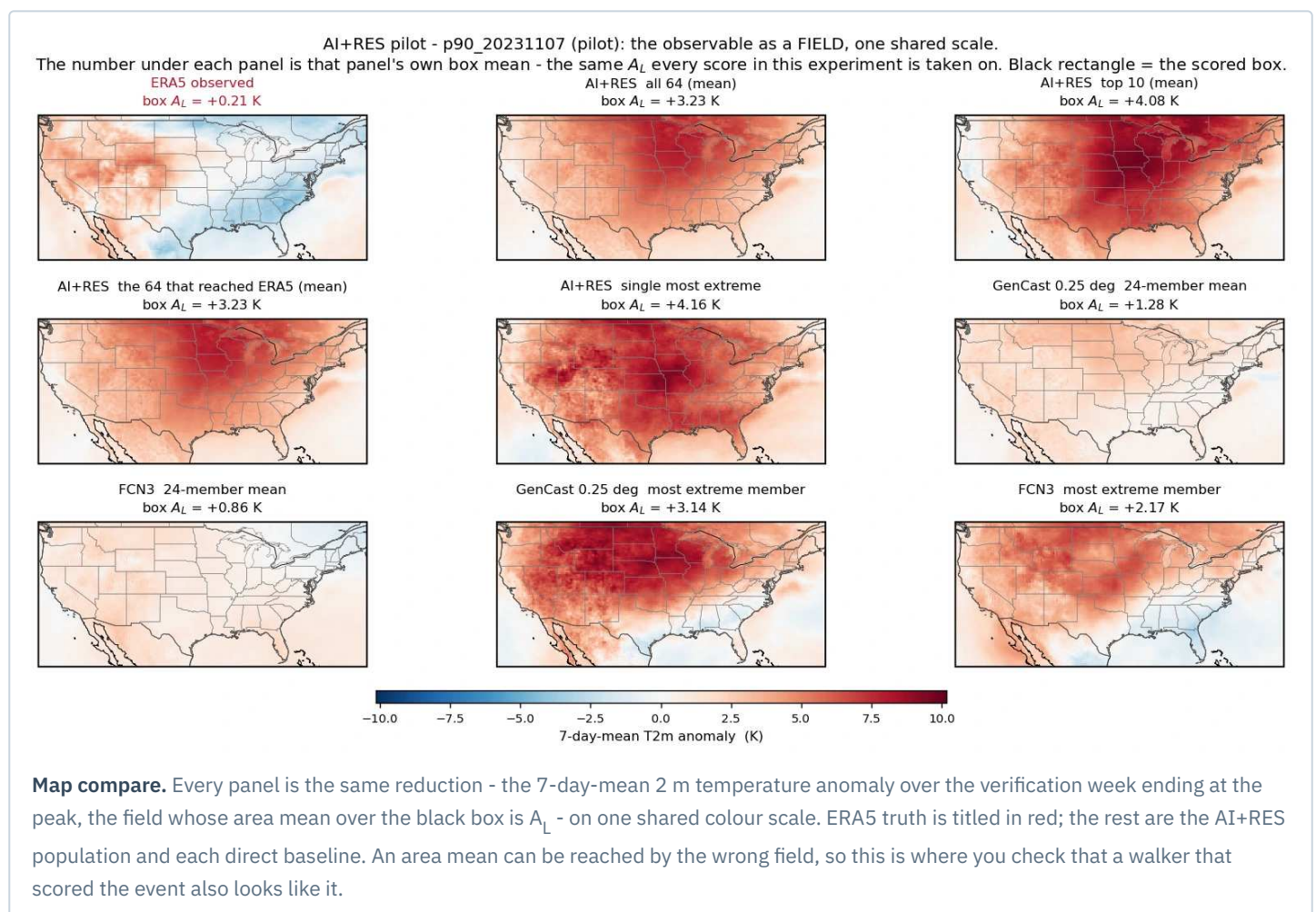
Every walker, and what it is worth. Top: all 64 surviving walkers ranked by how extreme they got, against the observed value in red and the best direct-sampling member in brown. Bottom: the probability mass $p_i = Z \cdot w_i / N$ each one carries, on a log axis, with the running total $\sum p_i$ as the black staircase - the headline probability is where that staircase stands when the ranking crosses the observation. Two reference lines set the scale: $1/n$, what a single direct member is worth, and $1/N$, what a walker would be worth if nothing had been resampled. Bars at identical height are clones of one parent at the last resampling and carry identical mass by construction, so the count of distinct masses under the panel - not 64 - is how many independent pieces of information the estimate rests on.

The estimator degenerates here, and finding out where that happens is the most useful thing the bottom of the ladder produced. Direct sampling puts this ordinary November week at 0.875; AI+RES reports 0.583, below the direct interval, and cannot report anything else.

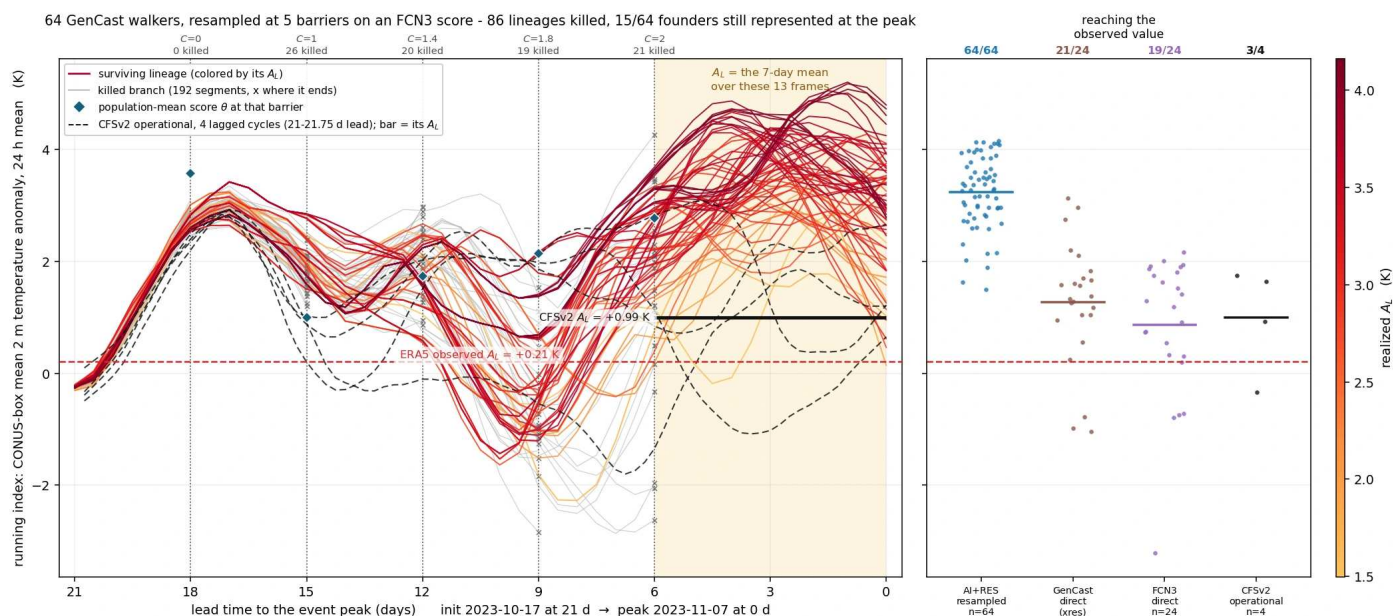
OBSERVED A_L +0.21 KERA5, 24-50 N, 125-66 W
CLIMATOLOGICAL PERCENTILE 60.10thERA5 1959-2022 · 53.64th against 1990-2019
RETURN PERIOD 1 yr62 of 64 years hold a window this extreme
DEPTH IN GENCAST'S SPREAD -0.97 σweek-3 ensemble +1.28 ± 1.09 K
DIRECT SAMPLING 21 / 24best member +3.14 K · P = 0.875 [0.676, 0.973]
AI+RES 64 / 64most extreme +4.16 K
WEIGHTED P(A_L BEYOND OBSERVED) 0.583 (pinned)sum of the walker probabilities past the observation
PEAK-DAY ANOMALY +3.01 Kthe single day, not the week · 98.7th percentile
64-YEAR RECORD FOR THIS SEASON +3.38 K (2020-11-10)peak 2023-11-07 · init 2023-10-17 (21 d lead)

box: Whole CONUS - this case is defined on the national mean (24-50 N, 125-66 W) · registry key p90_20231107 · Slurm job 1186, 7 h 25 m
run health: log Z -1.0156 · normalization check 0.5827 · 15/64 founders · ESS/N 1.00 / 0.41 / 0.62 / 0.53 / 0.53, the healthiest of the wave - which is the point: health metrics do not detect this failure

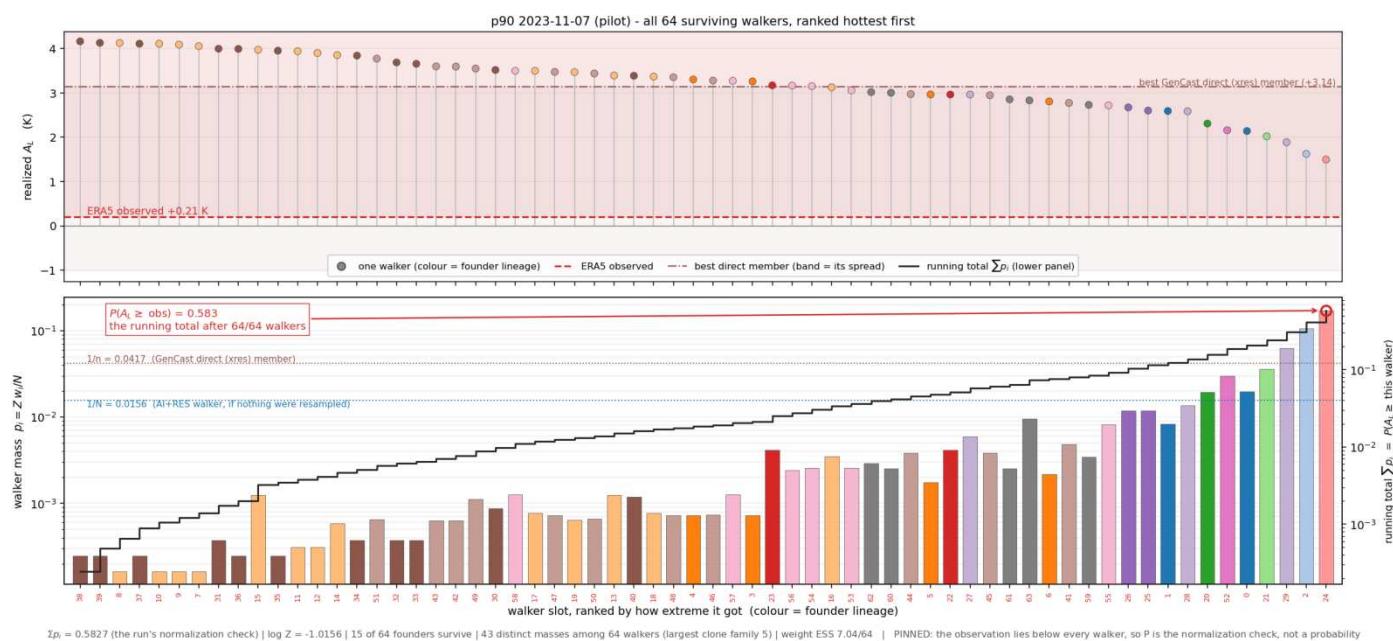
The cause is structural and was checked numerically rather than inferred. Resampling tilted the whole population warm, so its lowest realized A_L is +1.498 K - above the +0.211 K target - and the indicator $1\{A_L \geq \text{observed}\}$ is identically 1 across all 64 walkers. The estimate $Z \cdot \text{mean}(w \cdot 1\{\dots\})$ then collapses to $Z \cdot \text{mean}(w)$, which is exactly the normalization check: 0.582662 against 0.582662, to every digit printed. The estimator cannot return a probability above its own normalization check, and the truth sits above that ceiling. Self-normalizing is no better - it returns exactly 1.000, because “every walker reached it” is all the information the population retains. Nothing inside the run reveals any of this: a reader of compare.json alone would take 0.583 at face value. The rule it establishes is that AI+RES has no resolution at a target the tilt has already carried the entire population past, so the usable operating range is targets the resampled population still straddles - which the run one rung up does (its population minimum is +0.39 K against a +1.06 K target) and this one does not.



AI+RES pilot - p90_20231107 (pilot): the walk itself, barrier by barrier



Trajectory. The algorithm happening: A_L against lead time for every branch the run rolled, grey for lineages that were killed and coloured for the 64 that survived, with the resampling barriers as vertical lines, the 7-day verification window shaded, the observed value in red, and the operational CFSv2 forecast at the same lead in black.



Every walker, and what it is worth. Top: all 64 surviving walkers ranked by how extreme they got, against the observed value in red and the best direct-sampling member in brown. Bottom: the probability mass $p_i = Z \cdot w_i / N$ each one carries, on a log axis, with the running total $\sum p_i$ as the black staircase - the headline probability is where that staircase stands when the ranking crosses the observation. Two reference lines set the scale: $1/n$, what a single direct member is worth, and $1/N$, what a walker would be worth if nothing had been resampled. Bars at identical height are clones of one parent at the last resampling and carry identical mass by construction, so the count of distinct masses under the panel - not 64 - is how many independent pieces of information the estimate rests on.

The persistence control

same walkers, same schedule, different score

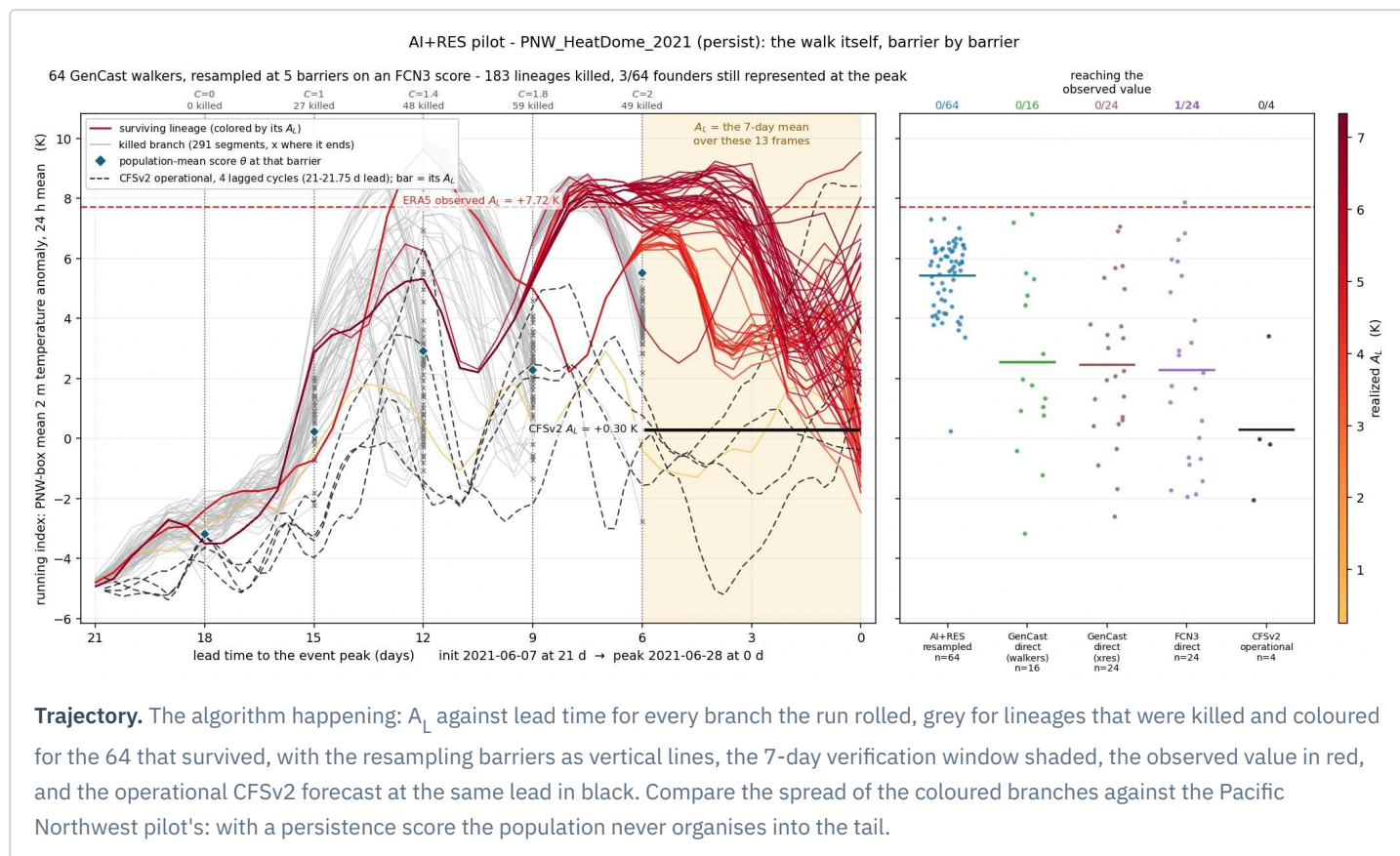
ATTRIBUTION

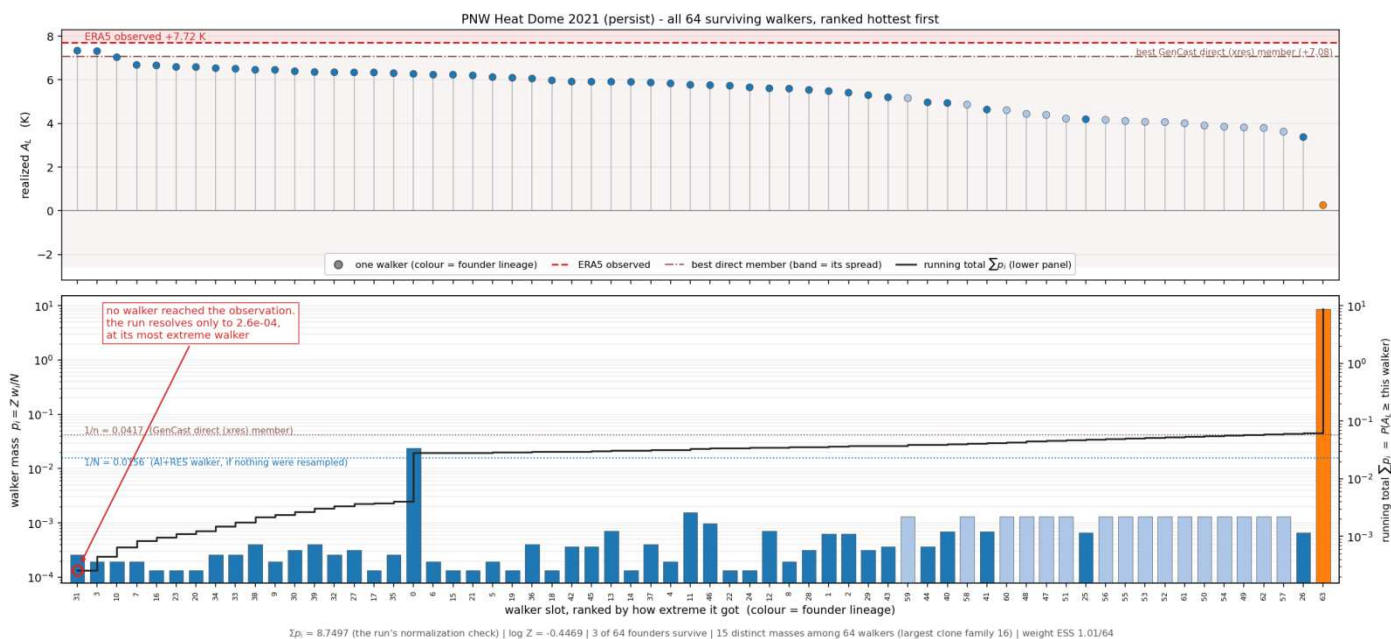
not reached

Every gain above is attributed to the FCN3 score by this one run. It repeats the Pacific Northwest pilot with the same walkers, the same resampling, the same C schedule and the same seeds, swapping only the function that decides which walkers to clone - from a skilful forecast of the event to “who is hottest now”. It reaches the observed +7.72 K with **0 of 64** walkers.

The selection is noise at this lead: ESS/N falls to 0.04 and 0.02 at steps 3 and 4 with clone multiplicities of 39 and 49, three of 64 founders survive, the weights span 0.013 to 869, and the normalization check comes out at 8.75 with a weighted mean of +0.29 K - the weights cancel the tilt entirely. Cloning on a skilful forecast of the event beats cloning on the present state, which is exactly what the Gate 3 persistence column predicted before any of these runs were made.

Slurm job 1180, 3 h 50 m · most extreme walker +7.33 K against an observed +7.72 K





Every walker, and what it is worth. The same panel as each event's, for the control. Nothing lies in the shaded band, so P is 0 and the honest statement is the deepest level the run resolved, 2.6×10^{-4} . Its 64 walkers hold only 15 distinct masses and a weight ESS of 1.0 - one effective sample - which is what a score with no skill at the target does to a population.

What is still missing

The figures are complete. All nine per-walker panels and the control's are on this page as of 26 August 2026 - the eight that were waiting on a3mega came across with the run reductions, and the cross-event view above replaces the one that was taken off for being unreadable.

The lift view. The other cross-event figure - the forecast scored against climatology rather than against the direct ensemble - is being rebuilt after it was caught scoring Winter Storm Elliott's Northern Plains index against the CONUS climatology column, which reported a 285× lift where the honest number is 0.35×. It is not on this page until it has been redrawn over the full slate.

One optional run. A 32-walker repeat of the 2 August 2024 normal-day test, to see whether the no-harm result survives half the population. Nothing on this page depends on it.

Where the method's limits now sit

Two boundaries are measured rather than assumed, and both were found by runs that did not go the way the plan expected. The **upper** limit is between +2.8 and +3.3 σ : Winter Storm Elliott at +2.81 σ is reached and the Southwest heat wave at +3.3 σ is not, and because one is a freeze and one a heat wave the boundary is not sign-dependent. The **lower** limit is a target the tilt has already carried the whole population past, where the estimate pins to the run's own normalization check and reports a probability that is silently meaningless. The method's usable range is the span between them: targets the resampled population still straddles, and that direct sampling still misses.

Provenance

Climatology from `python -m aires.aclim --report`, run on Derecho on 26 August 2026 against a freshly built cache: WeatherBench-2 daily-mean 0.25° ERA5 for 1959-2023, topped up from ARCO for the post-2023 peaks, anomalies against the same 1990-2019 baseline the experiment uses everywhere else. Return periods are a GEV fit to seasonal block maxima and are the softest number on the page - read them as an order of magnitude, not a figure.

Run statistics from each production's entry in `aires/HANDOFF.md`; figures from `aires/amaps.py` and `aires/aplots.py` on branch `aires`. Wave 1 - the four heat events and the persistence control - cost about 232 H100-hours across five runs and a gate; wave 2's five runs were budgeted at 61 to 64 H100-hours each.

A note on the two names without one: the `p90_*` cases are days selected by a national-mean anomaly threshold and carry no meteorological name, so they are named here by what they are - a nationwide warm spell at the 99th percentile, a normal-day test at the ensemble median, and a below-median day.